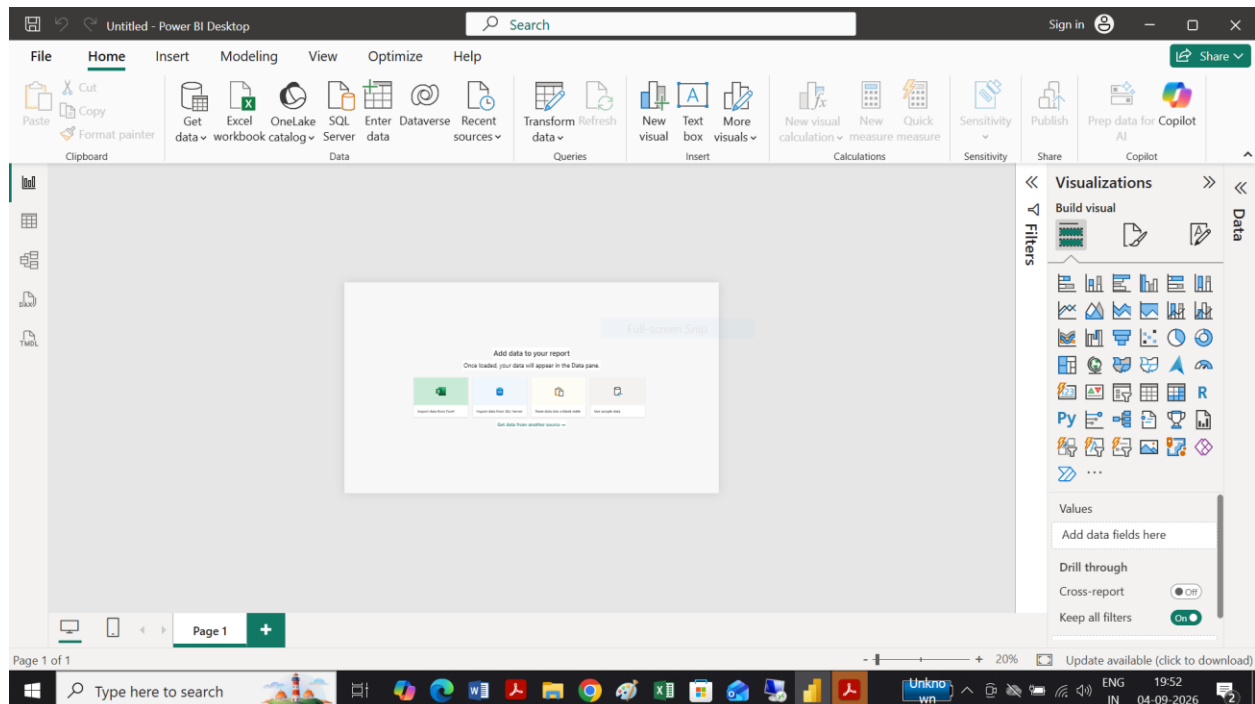


Session 1:

Tasks 1:



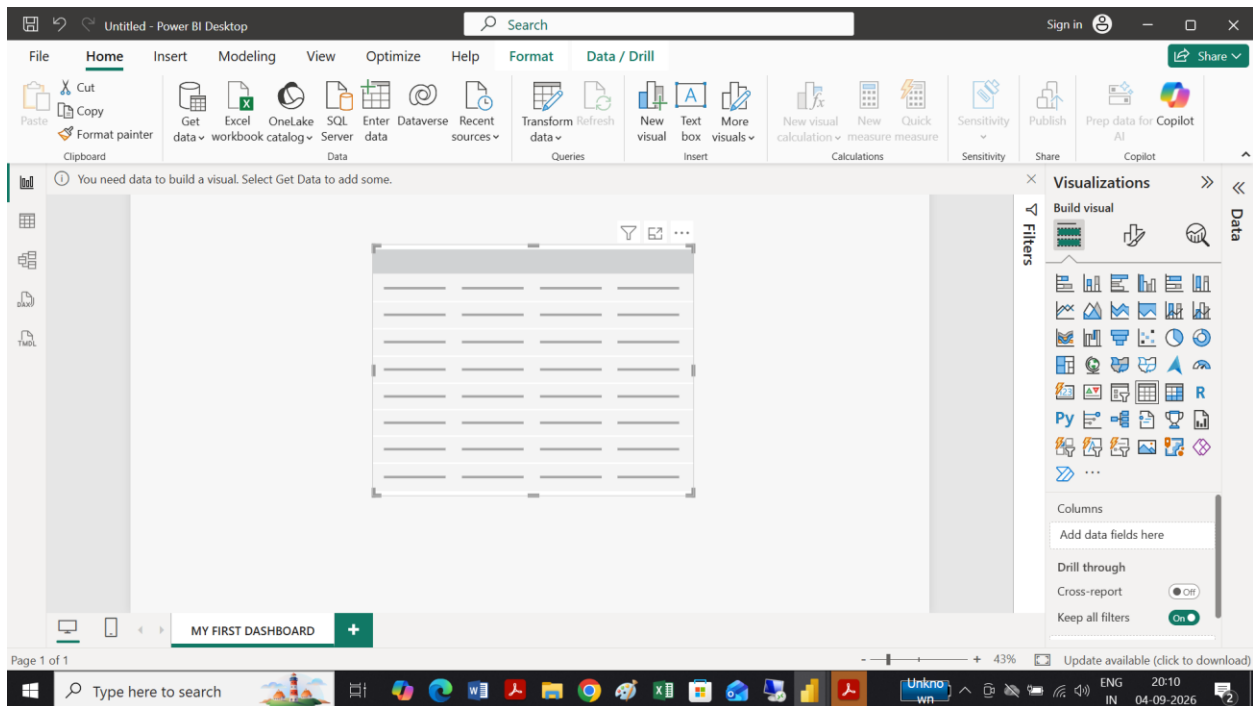
Tasks 2:

Power BI Desktop Interface Components

Component	Short Description
1. Report View	Used to create and design interactive reports and dashboards. You can add charts, tables, cards, slicers, and other visuals here.
2. Data View	Displays the actual data stored in your tables. It allows you to inspect, sort, and work with individual rows and columns of data.
3. Model View	Shows the relationships between different tables in the data model. It is used to create and manage relationships between tables.
4. Fields Pane	Lists all the tables, columns, and measures available in the data model. You can drag these fields onto visuals to build reports.

Component	Short Description
5. Visualizations Pane	Contains tools for creating and formatting visualizations such as bar charts, pie charts, tables, and graphs. It also provides formatting options for selected visuals.
6. Filters Pane	Allows you to filter the data displayed in a report or visual. Filters can be applied to a specific visual, an entire page, or the whole report

Tasks 3:



Excel Dashboard	Power BI Dashboard
1. Data & Analysis: Mainly uses worksheets, formulas, PivotTables, and PivotCharts for analysis.	1. Data & Analysis: Designed for connecting, transforming, and analyzing data from multiple sources using Power Query and DAX.
2. Visualization: Uses charts, tables, slicers, and other Excel	2. Visualization: Provides interactive visuals, filters, slicers,

Tasks 4:	objects that are arranged manually on a worksheet.	and a dedicated Visualizations pane for creating and formatting reports.	
Here are 3	3. Workflow & Sharing: Dashboards are generally maintained within Excel workbooks and shared as files.	3. Workflow & Sharing: Reports can be published to Power BI Service, where dashboards can be shared and viewed interactively across devices.	key

differences between Excel Dashboards and PowerBI Dashboards:

Session 2:

Tasks 1:

Report View:

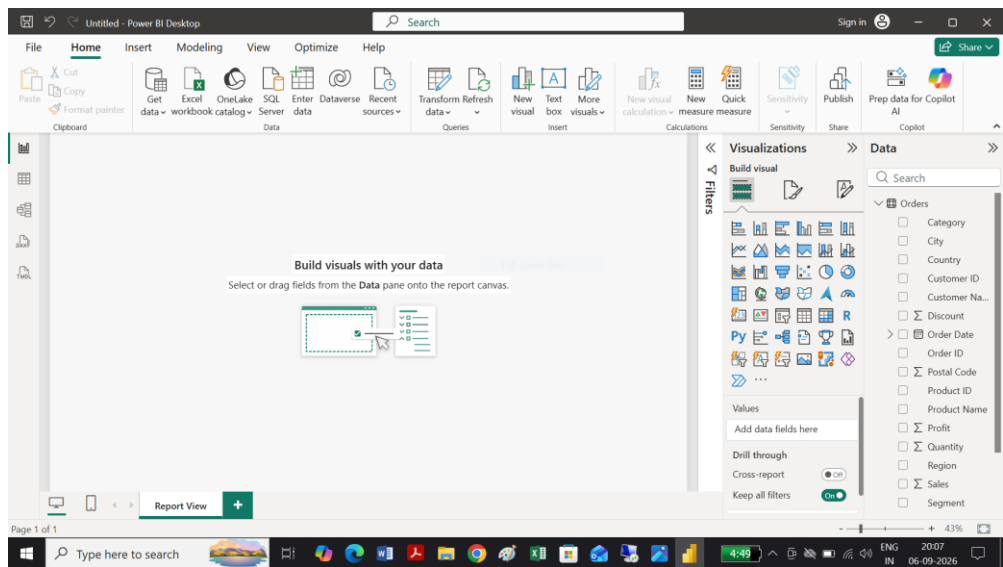


Table View:

Untitled - Power BI Desktop

File Home Help Table tools

Name Orders

Structure Relationships

Manage relationships Relationships

New measure measure column Calculations

Quick New table

Mark as date table Calendars

Data

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
CA-2016-101343	17 July 2016	22 July 2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	
CA-2017-163405	21 December 2017	25 December 2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	
CA-2017-163405	21 December 2017	25 December 2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	
US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	
US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	
US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	
US-2016-116729	25 December 2016	28 December 2016	First Class	GK-14620	Grace Kelly	Corporate	United States	Los Angeles	California	
CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	
CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	
CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	
CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	
CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	
CA-2016-118913	25 June 2016	29 June 2016	Standard Class	AS-10240	Alan Shonely	Consumer	United States	Los Angeles	California	
CA-2014-113579	13 December 2014	15 December 2014	Second Class	KD-16345	Katherine Duchich	Consumer	United States	Los Angeles	California	
CA-2017-126242	19 November 2017	24 November 2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California	
CA-2014-110849	18 April 2014	23 April 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	
CA-2014-110849	18 April 2014	23 April 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	
CA-2014-110849	18 April 2014	23 April 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	
CA-2016-130029	03 July 2016	06 July 2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	
CA-2016-130029	03 July 2016	06 July 2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	

Table: Orders (3,994 rows)

Model View:

Untitled - Power BI Desktop

File Home Help

Clipboard Data Queries Relationships Calculations Parameters Security Q&A Language Linguistic schema Sensitivity Publish

Get data Excel workbook OneLake catalog Enter data Dataverse SQL Server Recent sources

Transform Refresh data

Manage relationships Relationships

New measure column Calculations

New table

New calculation group

New parameter Parameters

Manage View roles as

Q&A setup

Language Linguistic schema

Sensitivity

Publish

Properties

Tables Model

Search

Orders

Category

City

Country

Customer ID

Customer Name

Discount

Order Date

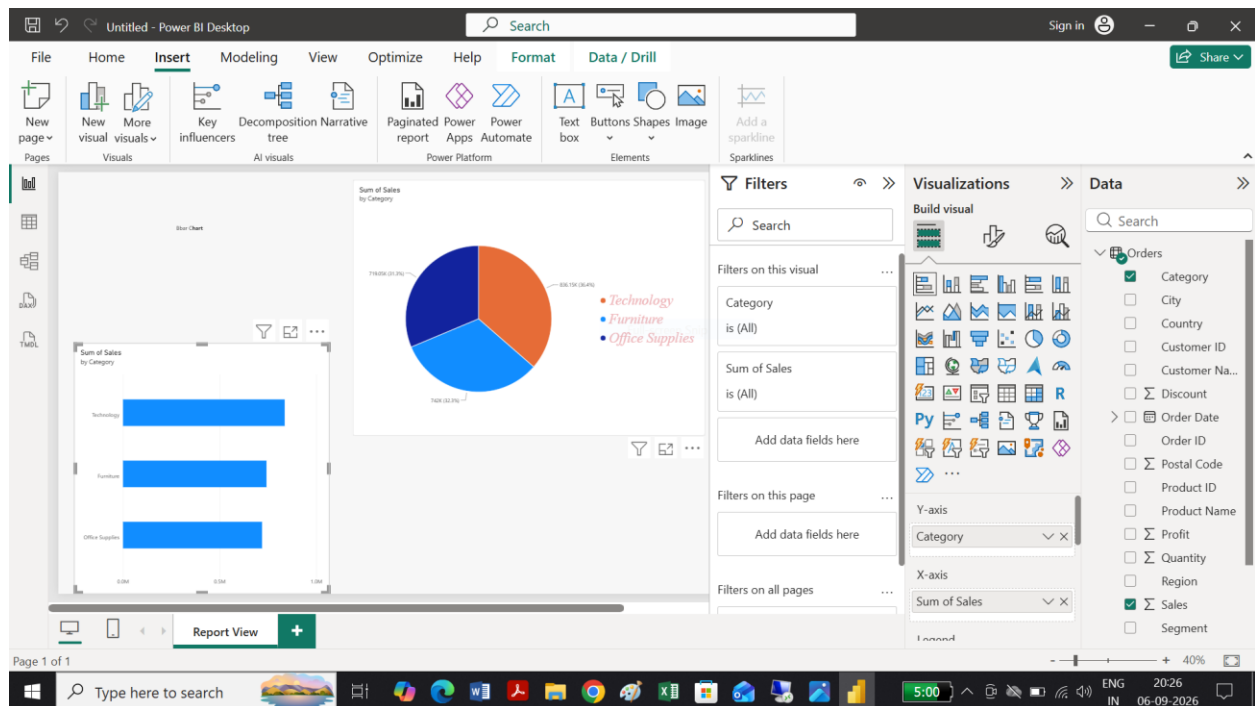
Order ID

Postal Code

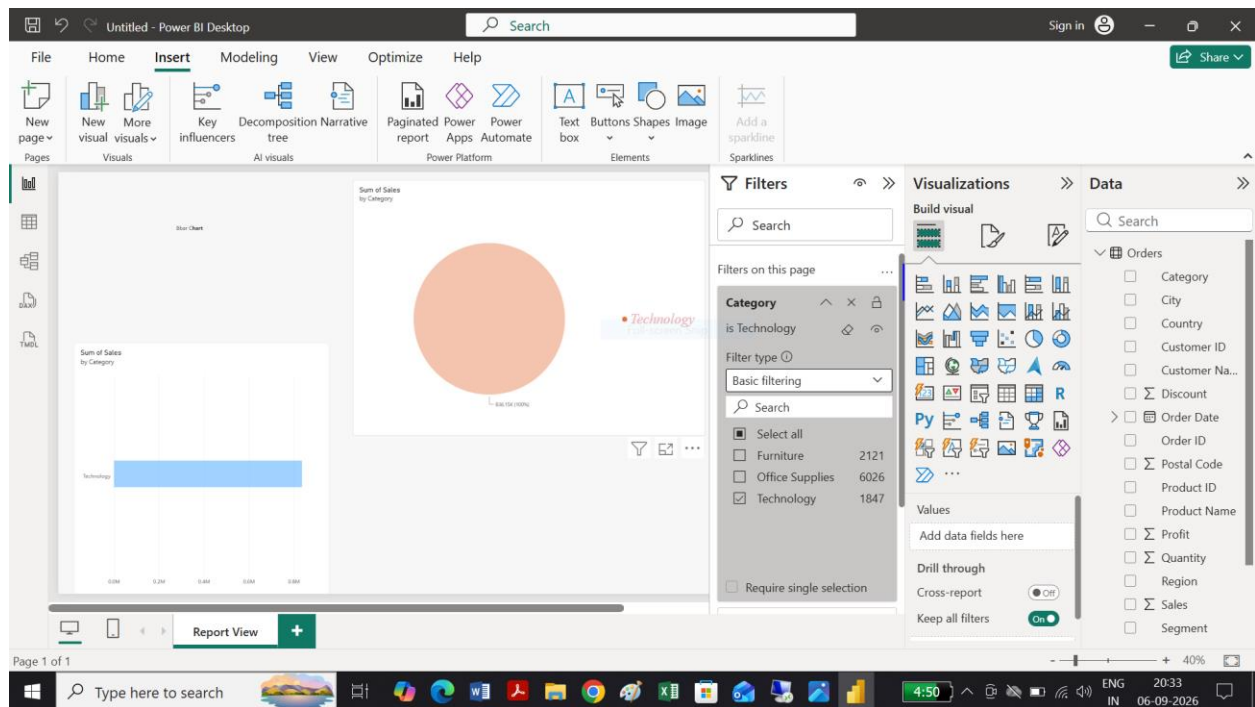
Full-screen Story

100%

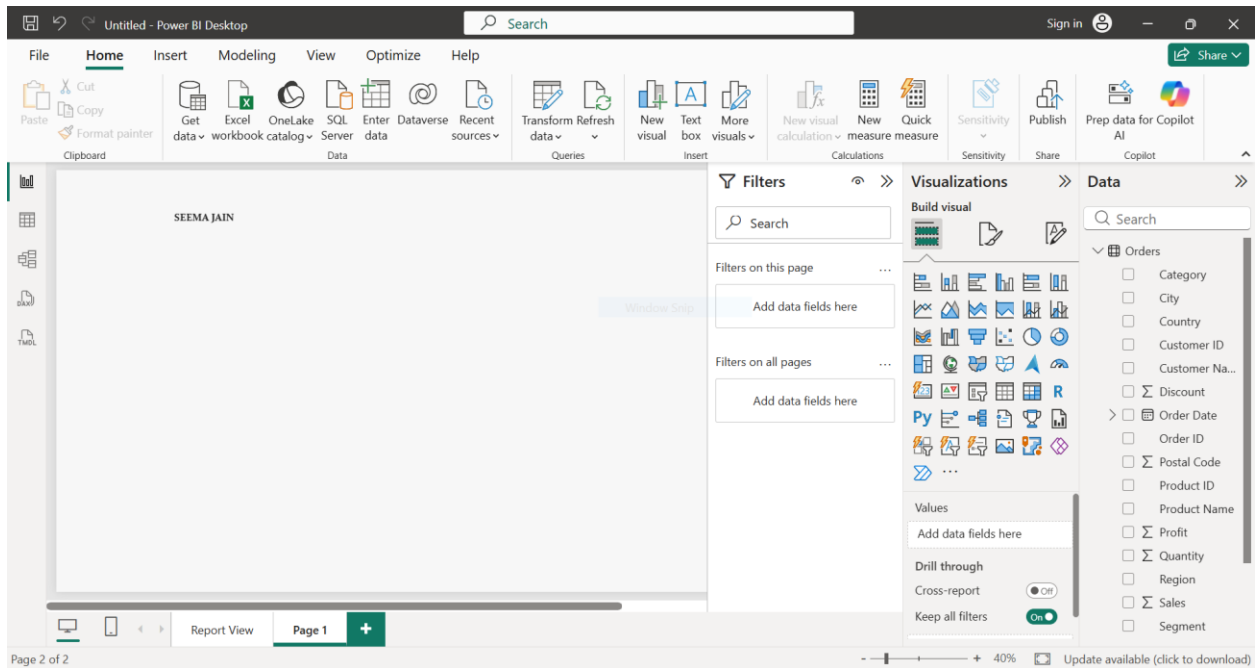
Task 2: BAR CHART & PIE CHART



Task 3: Sales Data for Single Category filter Used

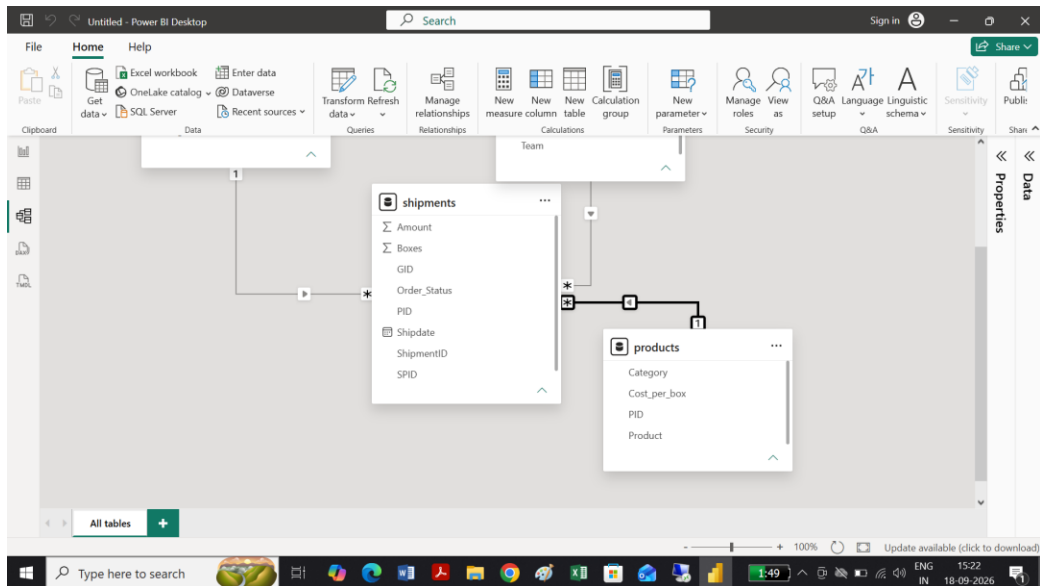


Task 4:



Task 5:

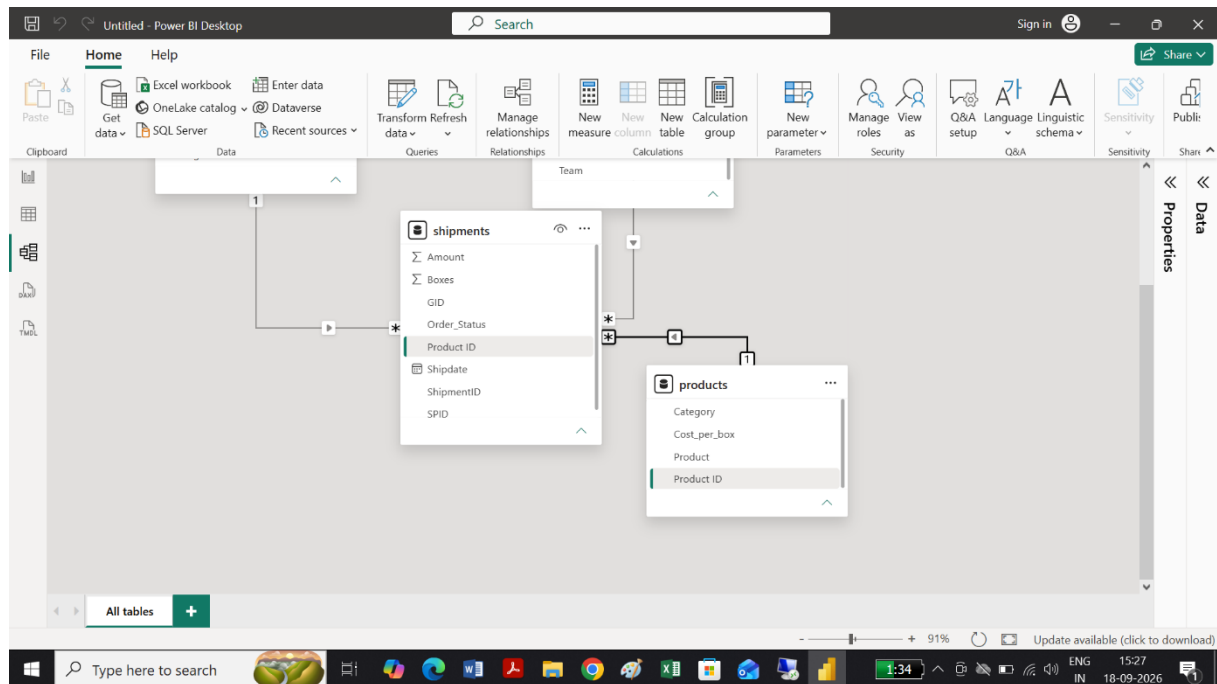
Relationship seen in model view between Shipments & Products and PID is renamed to Product ID as given below:



The screenshot shows the Power BI Desktop interface with a data table. The table has columns: ShipmentID, SPID, PID, GID, Shipdate, Amount, Boxes, and Order_Status. The table contains 22 rows of data. The 'PID' column is highlighted, and the 'Data' pane on the right shows the table structure. The interface includes a ribbon with tabs like File, Home, and Help, and a search bar at the top.

ShipmentID	SPID	PID	GID	Shipdate	Amount	Boxes	Order_Status
S00000593	SP01	P10	G6	11 October 2023	1372.5	99	Delivered
S00002124	SP01	P10	G6	17 May 2023	920.25	66	Delivered
S00003637	SP01	P10	G6	05 February 2024	4432.5	317	Delivered
S00003722	SP01	P10	G6	22 February 2023	823.5	64	Delivered
S00005183	SP01	P10	G6	17 April 2024	2574	190	Delivered
S00008276	SP01	P10	G6	16 June 2023	452.25	33	Delivered
S00008292	SP01	P10	G6	25 July 2023	7341.75	490	Delivered
S00009808	SP01	P10	G6	29 August 2023	1017	64	Delivered
S00009823	SP01	P10	G6	13 February 2023	2711.25	181	Delivered
S00011354	SP01	P10	G6	23 January 2023	18117	1295	Delivered
S00012849	SP01	P10	G6	14 March 2023	1293.75	87	Delivered
S00014346	SP01	P10	G6	07 March 2023	3411	244	Delivered
S00017458	SP02	P10	G6	28 November 2023	3645	243	Delivered
S00018988	SP02	P10	G6	13 May 2024	429.75	31	Delivered
S00019019	SP02	P10	G6	05 April 2023	562.5	47	Delivered
S00020532	SP02	P10	G6	08 February 2023	4497.75	346	Delivered
S00023525	SP02	P10	G6	25 December 2023	4671	334	Delivered
S00028134	SP03	P10	G6	16 June 2023	1885.5	110	Delivered
S00029715	SP03	P10	G6	05 January 2024	6412.5	401	Delivered
S00029746	SP03	P10	G6	15 February 2024	7364.25	567	Delivered
S00031277	SP03	P10	G6	28 May 2024	3021.75	189	Delivered

PID Renamed to Product Id



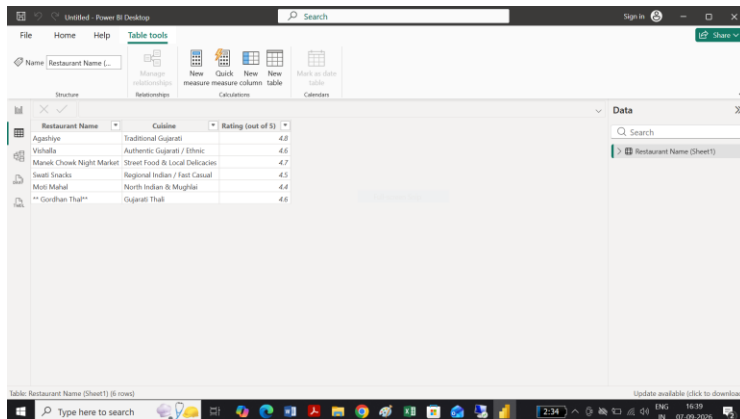
Session 3:

Tasks 1: Using Above data task 1 query is solved



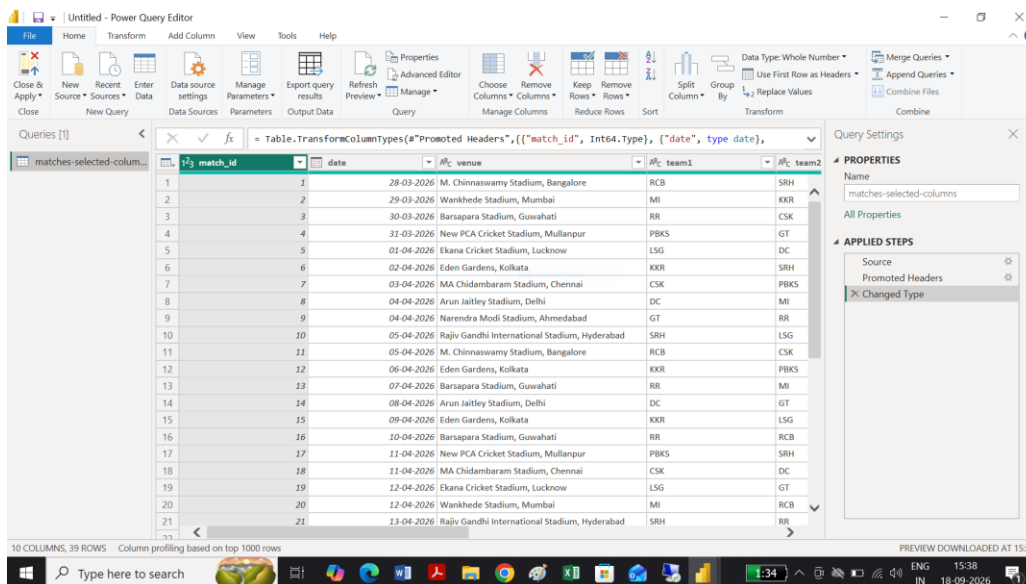
Favourite Restuarant
- (Name, Favouriities &

Data loaded in Power -Bi as such seen in Table view screenshot below



Tasks 2:

Direct Query by Transforming data



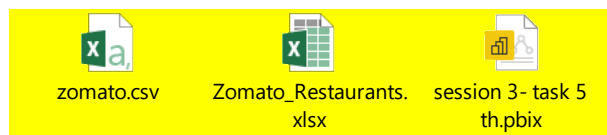
One-line difference: **Import** copies the data into Power BI for faster analysis, while **DirectQuery** leaves the data in the source and queries it whenever visuals need data.

So, for the **provided CSV**, use **Import mode**; you cannot directly select **DirectQuery** for this file.

Tasks 3: Pending

Task 4: Connect Power bi to SQL : Pending

Task 5: Using below data task 5 query is solved



The image displays a screenshot of the Microsoft Power BI Desktop interface. The top section shows the 'Manage relationships' window, which is used to define data relationships between tables. It lists a relationship between 'zomato (Restaurant Name)' and 'Zomato_Restaurants (Restaura...)' with a cardinality of 1 to 1 and an active status.

The main workspace shows a visual titled 'Average of Average Cost for item: Locality'. The visual displays a table with two columns: 'Locality' and 'Average Cost'. The data is as follows:

Locality	Average Cost
002.00 - A-Hotel, Gurdev Nagar	800.00

The right-hand pane shows the 'Visualizations' pane with various chart types and the 'Data' pane showing the fields available for the visual. The 'Data' pane lists fields such as Address, Aggregate rat..., Average Cost, City, Country Code, Cuisines, Currency, Has Online de..., Has Table boo..., Is delivering n..., Latitude, Locality Verbo..., Longitude, Price range, and Rating color. The 'Locality' field is selected for the visual.

The bottom of the screen shows the Windows taskbar with the time 1:50 and date 08-09-2026.

Session 4:

Tasks 1:

POWE BI
ASSIGNMENT.pbix

matches-selected-col
umns.csv

POWE BI ASSIGNMENT

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1] matches-selected-column...

Table.TransformColumnTypes(#"Promoted Headers",{"match_id", Int64.Type}, {"date", type date},

match_id date venue team1 team2

1 1 28-03-2026 M. Chinnaswamy Stadium, Bangalore RCB SRH

2 2 29-03-2026 Wankhede Stadium, Mumbai MI KKR

3 3 30-03-2026 Barsapara Stadium, Guwahati RR CSK

4 4 31-03-2026 New PCA Cricket Stadium, Mullanpur PBKS GT

5 5 01-04-2026 Ekana Cricket Stadium, Lucknow LSG DC

6 6 02-04-2026 Eden Gardens, Kolkata KKR SRH

7 7 03-04-2026 MA Chidambaram Stadium, Chennai CSK PBKS

8 8 04-04-2026 Arun Jaitley Stadium, Delhi DC MI

9 9 04-04-2026 Narendra Modi Stadium, Ahmedabad GT RR

10 10 05-04-2026 Rajiv Gandhi International Stadium, Hyderabad SRH LSG

11 11 05-04-2026 M. Chinnaswamy Stadium, Bangalore RCB CSK

12 12 06-04-2026 Eden Gardens, Kolkata KKR PBKS

13 13 07-04-2026 Barsapara Stadium, Guwahati RR MI

14 14 08-04-2026 Arun Jaitley Stadium, Delhi DC GT

15 15 09-04-2026 Eden Gardens, Kolkata KKR LSG

16 16 10-04-2026 Barsapara Stadium, Guwahati RR RCB

17 17 11-04-2026 New PCA Cricket Stadium, Mullanpur PBKS SRH

18 18 11-04-2026 MA Chidambaram Stadium, Chennai CSK DC

19 19 12-04-2026 Ekana Cricket Stadium, Lucknow LSG GT

20 20 12-04-2026 Wankhede Stadium, Mumbai MI RCB

21 21 13-04-2026 Rajiv Gandhi International Stadium, Hyderabad SRH RR

10 COLUMNS, 39 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 16:56

2:29 17:03 07-09-2026

Tasks 2:

POWE BI
ASSIGNMENT- 2.pbix

POWE BI ASSIGNMENT- 2

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Merge Queries Append Queries Combine Files Combine

Queries [2] matches-selected-column...

Table.RemoveColumns(*"Changed Type",{"loss_decision", "stage"})

match_id date venue team1 team2

1 1 28-03-2026 M. Chinnaswamy Stadium, Bangalore RCB SRH

2 2 29-03-2026 Wankhede Stadium, Mumbai MI KKR

3 3 30-03-2026 Barsapara Stadium, Guwahati RR CSK

4 4 31-03-2026 New PCA Cricket Stadium, Mullanpur PBKS GT

5 5 01-04-2026 Ekana Cricket Stadium, Lucknow LSG DC

6 6 02-04-2026 Eden Gardens, Kolkata KKR SRH

7 7 03-04-2026 MA Chidambaram Stadium, Chennai CSK PBKS

8 8 04-04-2026 Arun Jaitley Stadium, Delhi DC MI

9 9 04-04-2026 Narendra Modi Stadium, Ahmedabad GT RR

10 10 05-04-2026 Rajiv Gandhi International Stadium, Hyderabad SRH LSG

11 11 05-04-2026 M. Chinnaswamy Stadium, Bangalore RCB CSK

12 12 06-04-2026 Eden Gardens, Kolkata KKR PBKS

13 13 07-04-2026 Barsapara Stadium, Guwahati RR MI

14 14 08-04-2026 Arun Jaitley Stadium, Delhi DC GT

15 15 09-04-2026 Eden Gardens, Kolkata KKR LSG

16 16 10-04-2026 Barsapara Stadium, Guwahati RR RCB

17 17 11-04-2026 New PCA Cricket Stadium, Mullanpur PBKS SRH

18 18 11-04-2026 MA Chidambaram Stadium, Chennai CSK DC

19 19 12-04-2026 Ekana Cricket Stadium, Lucknow LSG GT

20 20 12-04-2026 Wankhede Stadium, Mumbai MI RCB

21 21 13-04-2026 Rajiv Gandhi International Stadium, Hyderabad SRH RR

10 COLUMNS, 39 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 17:10

2:29 17:10 07-09-2026

Task 3:

POWE BI ASSIGNMENT- 2

File Home Transform Add Column View Tools Help

Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Split Column, Group By, Data Type: Whole Number, Use First Row as Headers, Merge Queries, Append Queries, Combine Files

Queries [1] = Table.Skip(#"Removed Blank Rows",2)

match_id	date	venue	team1	team2
1	30-03-2026	Barsapara Stadium, Guwahati	RR	CSK
2	31-03-2026	New PCA Cricket Stadium, Mullanpur	PBKS	GT
3	01-04-2026	Ekana Cricket Stadium, Lucknow	LSG	DC
4	02-04-2026	Eden Gardens, Kolkata	KKR	SRH
5	03-04-2026	MA Chidambaram Stadium, Chennai	CSK	PBKS
6	04-04-2026	Arun Jaitley Stadium, Delhi	DC	MI
7	04-04-2026	Narendra Modi Stadium, Ahmedabad	GT	RR
8	05-04-2026	Rajiv Gandhi International Stadium, Hyderabad	SRH	LSG
9	05-04-2026	M. Chinnaswamy Stadium, Bangalore	RCB	CSK
10	06-04-2026	Eden Gardens, Kolkata	KKR	PBKS
11	07-04-2026	Barsapara Stadium, Guwahati	RR	MI
12	08-04-2026	Arun Jaitley Stadium, Delhi	DC	GT
13	09-04-2026	Eden Gardens, Kolkata	KKR	LSG
14	10-04-2026	Barsapara Stadium, Guwahati	RR	RCB
15	11-04-2026	New PCA Cricket Stadium, Mullanpur	PBKS	SRH
16	11-04-2026	MA Chidambaram Stadium, Chennai	CSK	DC
17	12-04-2026	Ekana Cricket Stadium, Lucknow	LSG	GT
18	12-04-2026	Wankhede Stadium, Mumbai	MI	RCB
19	13-04-2026	Rajiv Gandhi International Stadium, Hyderabad	SRH	RR
20	14-04-2026	MA Chidambaram Stadium, Chennai	CSK	KKR
21	15-04-2026	M. Chinnaswamy Stadium, Bangalore	RCB	LSG

10 COLUMNS, 37 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 17:35

Query Settings

PROPERTIES

Name: matches-selected-columns

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Blank Rows
- Filtered Rows
- Filtered Rows1
- Filtered Rows2

Task 4:

POWE BI ASSIGNMENT- 2

File Home Transform Add Column View Tools Help

Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Split Column, Group By, Data Type: Text, Use First Row as Headers, Merge Queries, Append Queries, Combine Files

Queries [1] = Table.Distinct(#"Removed Duplicates1",{"team2"})

date	venue	team1	team2
28-03-2026	M. Chinnaswamy Stadium, Bangalore	RCB	SRH
29-03-2026	Wankhede Stadium, Mumbai	MI	KKR
30-03-2026	Barsapara Stadium, Guwahati	RR	CSK
31-03-2026	New PCA Cricket Stadium, Mullanpur	PBKS	GT
01-04-2026	Ekana Cricket Stadium, Lucknow	LSG	DC
03-04-2026	MA Chidambaram Stadium, Chennai	CSK	PBKS
04-04-2026	Arun Jaitley Stadium, Delhi	DC	MI
04-04-2026	Narendra Modi Stadium, Ahmedabad	GT	RR
05-04-2026	Rajiv Gandhi International Stadium, Hyderabad	SRH	LSG

10 COLUMNS, 9 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 18:02

Query Settings

PROPERTIES

Name: matches-selected-columns

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Removed Duplicates1
- Removed Duplicates2

Task 5:

POWE BI ASSIGNMENT- 2

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows Table

Transpose Reverse Rows Detect Data Type Rename Replace Values Fill Pivot Column Unpivot Columns Move Convert to List Split Column Format Text Column Merge Columns Extract Parse Statistics Standard Scientific Rounding Information

Date Time Duration Run R script Run Python script

Query Settings

PROPERTIES

Name matches-selected-columns

All Properties

APPLIED STEPS

Source Replaced Errors

Table: Table.ReplaceErrorValues(Source, {"Column7", "Unknown"})

mn3	Column4	Column5	Column6	Column7	Column8
1	team1	team2	stage	toss_winner	toss_decision
2	aswamy Stadium, Bangalore	RCB	SRH	League	RCB
3	Je Stadium, Mumbai	MI	KKR	League	MI
4	a Stadium, Guwahati	RR	CSK	League	RR
5	Cricket Stadium, Mullanpur	PBKS	GT	League	PBKS
6	icket Stadium, Lucknow	LSG	DC	League	DC
7	dens, Kolkata	KKR	SRH	League	KKR
8	ambaram Stadium, Chennai	CSK	PBKS	League	PBKS
9	ey Stadium, Delhi	DC	MI	League	DC
10	Modi Stadium, Ahmedabad	GT	RR	League	RR
11	dhi International Stadium, Hyderabad	SRH	LSG	League	LSG
12	aswamy Stadium, Bangalore	RCB	CSK	League	CSK
13	dens, Kolkata	KKR	PBKS	League	KKR
14	a Stadium, Guwahati	RR	MI	League	MI
15	ey Stadium, Delhi	DC	GT	League	DC
16	dens, Kolkata	KKR	LSG	League	LSG
17	a Stadium, Guwahati	RR	RCB	League	RR
18	Cricket Stadium, Mullanpur	PBKS	SRH	League	PBKS
19	ambaram Stadium, Chennai	CSK	DC	League	DC
20	icket Stadium, Lucknow	LSG	GT	League	GT
21	Je Stadium, Mumbai	MI	RCB	League	MI

10 COLUMNS, 40 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 18:09

1:51 ENG IN 18:11 07-09-2026

Session 5:

Tasks 1:



zomato_listings.csv

Using Above data task 1 query is solved

Untitled - Power BI Desktop

File Home Help

Table tools

Column tools

Name | format_settings

Structure

Columns

Manage relationships

New measure

Quick New

New measure

Measure

Columns

Relationships

Calculations

Start an online help

Columns

Data

Search

format_settings

Restaurant ID	Restaurant Name	Area	Cuisine	Average Cost for Two	Rating	Votes	Columns
101	Baham's - Oldemans Enclave	North Indian		3500	4.6	4510	
102	Tuffies - Koramangala	Burgers & American		900	4.5	8910	
103	Saravana Bhavan - Connaught Place	South Indian		500	4.2	3210	
104	Pan Cut - Park Street	Continental & Mughlai		1200	4.4	6540	
105	Karavalli - Residency Road	Coastal & South Indian		2500	4.5	2100	
106	Bademiya - Cobalt	Street Food & Kabab		700	4.1	5430	
107	Glen's Bakeshops - Indiranagar	Bakery & Desserts		800	4.3	3890	
108	Pandora's - Seashore Road	Biryani & Hyderabad		800	4.2	7830	
109	Tot - Indiranagar	Italian & Microbrewery		1800	4.6	9210	
110	Brianco's & Co. - Fort	Parisi & Italian		1000	4.2	3100	
111	Anchor Brewing Company - Adamb Nagar	American & Microbrewery		2000	4.4	3600	
112	Nagaraja - MG Road	Andhra & South Indian		800	4.3	4300	
113	Indian Coffee House - College Street	Snacks & South Indian		400	4.0	1800	
114	MTR - Lullaby	South Indian		400	4.4	1100	
115	Bhann Sen - Ischi - Sadashiva Pet	Maharashtrian		500	4.2	1200	

Table format_settings [1] view Columns Columns [1 distinct values]

Type here to search

Update available (click to download)

Task 2:



Flipkart_User_Data.xl
SX

Using Above data task 1 query is solved



Tasks 3:



Swiggy_Orders_Data
set.xlsx

Using Above data task 1 query is solved

Order ID	Item Name	Status	Location	Order Type	Order Status	Order Date
0807013	100	Cancelled	400	Bangalore	Credit Card	08/07/2013
0807012	100	Cancelled	400	Bangalore	Credit Card	08/07/2012
0807010	100	Cancelled	400	Bangalore	Credit Card	08/07/2010
0807007	100	Cancelled	400	Bangalore	Credit Card	08/07/2007
0807004	100	Cancelled	400	Bangalore	Credit Card	08/07/2004
0807002	100	Cancelled	400	Bangalore	Credit Card	08/07/2002

Task 4:

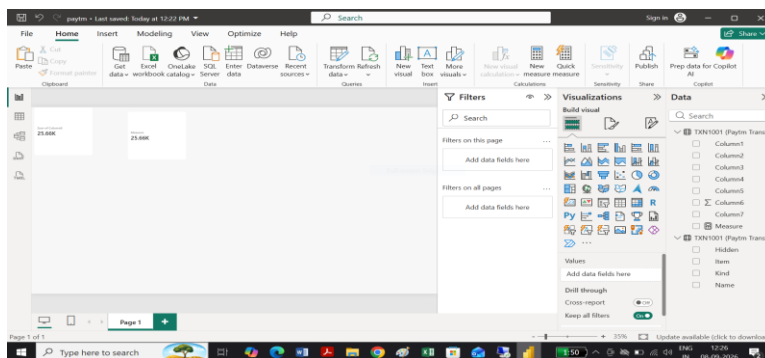


Paytm_Transaction_D
ataset.xlsx

Using Above data task 1 query is solved

Transaction ID	Date	Time	Category	Item	Status	Amount
TAN1001	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	25000
TAN1002	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	15000
TAN1003	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	45000
TAN1004	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	80000
TAN1005	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	210000
TAN1006	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	45000
TAN1007	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	80000
TAN1008	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	25000
TAN1009	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	15000
TAN1010	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	45000
TAN1011	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	80000
TAN1012	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	210000
TAN1013	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	45000
TAN1014	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	80000
TAN1015	2024-09-01	10:30:00	Payment Method	Bank Transfer & Bk	Success	25000

And as seen below sum of column by graph and measure is same as per the query request the numeric value required



Task 5:



Movie_Booking_Data
set_FillDown_Practice

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1] `= Table.FillDown(#"Replaced Value",{"Movie Name"})`

	Booking ID	Movie Name	Show Date	Show Time	Theater Name
1	BK1001	Avatar: The Way of Water	01-10-2026	10:30:00	PVR IMAX Forum
2	BK1002	Avatar: The Way of Water	01-10-2026	13:45:00	PVR IMAX Forum
3	BK1003	Avatar: The Way of Water	01-10-2026	17:15:00	PVR IMAX Forum
4	BK1004	Avatar: The Way of Water	01-10-2026	20:45:00	PVR IMAX Forum
5	BK1005	Oppenheimer	01-10-2026	11:00:00	INX Multiplex
6	BK1006	Oppenheimer	01-10-2026	14:30:00	INX Multiplex
7	BK1007	Oppenheimer	01-10-2026	18:00:00	INX Multiplex
8	BK1008	Spider-Man: Across the Spider-Verse	02-10-2026	09:15:00	Cinepolis Grand
9	BK1009	Spider-Man: Across the Spider-Verse	02-10-2026	12:30:00	Cinepolis Grand
10	BK1010	Spider-Man: Across the Spider-Verse	02-10-2026	16:00:00	Cinepolis Grand
11	BK1011	Spider-Man: Across the Spider-Verse	02-10-2026	19:30:00	Cinepolis Grand
12	BK1012	Jawan	02-10-2026	10:00:00	Miraj Cinemas
13	BK1013	Jawan	02-10-2026	13:15:00	Miraj Cinemas
14	BK1014	Jawan	02-10-2026	16:30:00	Miraj Cinemas
15	BK1015	Dune: Part Two	03-10-2026	11:45:00	PVR Gold Class
16	BK1016	Dune: Part Two	03-10-2026	15:15:00	PVR Gold Class
17	BK1017	Dune: Part Two	03-10-2026	19:00:00	PVR Gold Class

Query Settings

PROPERTIES

Name
Booking ID (Movie Booking Data)

All Properties

APPLIED STEPS

- Source
- Navigation
- FilterNullAndWhitespace
- Removed Bottom Rows
- Removed Top Rows
- Promoted Headers
- Changed Type
- Replaced Value
- Filled Down**

10 COLUMNS, 17 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 18:43

Same as above query solved seen in Table view below as seen:

Untitled - Power BI Desktop

File Home Help Table tools

Name: Booking ID (Movie ...)

Manage relationships

New measure Quick New New Mark as date

measure column table table

Calculations Calendars

Booking ID	Movie Name	Show Date	Show Time	Theater Name	Screen	Category	Tickets Booked	Ticket Price (₹)	Tot
BK1001	Avatar: The Way of Water	01 October 2026	10:30:00	PVR IMAX Forum	Screen 1	Recliner	2	450	
BK1002	Avatar: The Way of Water	01 October 2026	13:45:00	PVR IMAX Forum	Screen 1	Prime	4	300	
BK1003	Avatar: The Way of Water	01 October 2026	17:15:00	PVR IMAX Forum	Screen 1	Classic	1	200	
BK1004	Avatar: The Way of Water	01 October 2026	20:45:00	PVR IMAX Forum	Screen 1	Recliner	3	450	
BK1005	Oppenheimer	01 October 2026	11:00:00	INX Multiplex	Screen 3	IMAX 2D	2	500	
BK1006	Oppenheimer	01 October 2026	14:30:00	INX Multiplex	Screen 3	IMAX 2D	2	500	
BK1007	Oppenheimer	01 October 2026	18:00:00	INX Multiplex	Screen 3	Regular	5	250	
BK1008	Spider-Man: Across the Spider-Verse	02 October 2026	09:15:00	Cinepolis Grand	Screen 2	Executive	2	280	
BK1009	Spider-Man: Across the Spider-Verse	02 October 2026	12:30:00	Cinepolis Grand	Screen 2	VIP	2	600	
BK1010	Spider-Man: Across the Spider-Verse	02 October 2026	16:00:00	Cinepolis Grand	Screen 2	Executive	3	280	
BK1011	Spider-Man: Across the Spider-Verse	02 October 2026	19:30:00	Cinepolis Grand	Screen 2	VIP	4	600	
BK1012	Jawan	02 October 2026	10:00:00	Miraj Cinemas	Screen 4	Normal	1	180	
BK1013	Jawan	02 October 2026	13:15:00	Miraj Cinemas	Screen 4	Normal	2	180	
BK1014	Jawan	02 October 2026	16:30:00	Miraj Cinemas	Screen 4	Gold	4	320	
BK1015	Dune: Part Two	03 October 2026	11:45:00	PVR Gold Class	Screen 5	Luxury	2	800	
BK1016	Dune: Part Two	03 October 2026	15:15:00	PVR Gold Class	Screen 5	Luxury	2	800	
BK1017	Dune: Part Two	03 October 2026	19:00:00	PVR Gold Class	Screen 5	Luxury	6	800	

Table: Booking ID (Movie Booking Data) (17 rows)

Update available (click to download)

Session 6:

Task 1:

Using below data task 1 query is solved



food_delivery_orders.
xlsx

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Transpose Reverse Rows

Data Type: Any Replace Values Detect Data Type Fill Rename

Unpivot Columns Move Pivot Column Convert to List

Split Column Format Extract Parse

Merge Columns

Statistics Standard Scientific

Trigonometry Rounding Information

Date Time Duration

Run R script Run Python script

Queries [2]

Table

fx = Table.RenameColumns(#"Unpivoted Only Selected Columns",({"Attribute", "Month"}, {"Value", "Order Count"}))

Original Data

Food Delivery Orders - ...

Column1	Column2	Column3	Month	Order Count
1	OrderID	Restaurant	Column4	Month1
2	OrderID	Restaurant	Column5	Month2
3	OrderID	Restaurant	Column6	Month3
4				
5		101 Spice Villa	Column4	15
6		101 Spice Villa	Column5	18
7		101 Spice Villa	Column6	25
8		102 Pizza Heaven	Column4	20
9		102 Pizza Heaven	Column5	22
10		102 Pizza Heaven	Column6	28
11		103 Burger Hub	Column4	10
12		103 Burger Hub	Column5	15
13		103 Burger Hub	Column6	18
14		104 Tandoor Express	Column4	25
15		104 Tandoor Express	Column5	30
16		104 Tandoor Express	Column6	35
17		105 Noodle Bar	Column4	12
18		105 Noodle Bar	Column5	14
19		105 Noodle Bar	Column6	20
20		106 Taco Fiesta	Column4	18
21		106 Taco Fiesta	Column5	21
22		106 Taco Fiesta	Column6	25

5 COLUMNS, 33 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 08:32

Task 2:

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Transpose Reverse Rows

Data Type: Decimal Number Replace Values Detect Data Type Fill Rename

Unpivot Columns Move Pivot Column Convert to List

Split Column Format Extract Parse

Merge Columns

Statistics Standard Scientific

Trigonometry Rounding Information

Date Time Duration

Run R script Run Python script

Queries [1]

Table

fx = Table.Group(#"Changed Type", {"ProductCategory"}, ({"Total Sales Amount", each List.Sum([TotalsSales])

Sheet1

ProductCategory	Total Sales Amount	
1	Electronics	32000
2	Fashion	4400
3	Grocery	2400
4	Home Appliances	3500

2 COLUMNS, 4 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 22:11

Query Settings

PROPERTIES

Name

Sheet1

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Grouped Rows

The screenshot shows the Microsoft Power Query Editor interface. At the top, there's a ribbon with tabs: File, Home, Transform, Add Column, View, Tools, and Help. Below the ribbon are several tool groups:

- Columns**: Column from Custom Examples, Column Function, Invoke Custom, General.
- Merge**: Merge Columns.
- Conditional Column**: Conditional Column, Index Column, Duplicate Column.
- Format**: Format, Parse.
- Statistics**: Statistics, Standard Scientific.
- Trigonometry**: Trigonometry, Rounding, Information.
- Date**: Date, Time, Duration, From Date & Time.

The main workspace shows a query named "Table.AddColumn(#\"changed Type\", \"Artist - Song Title\", each [artist_name] & \"-\" & [track_name])". Below the formula bar, there's a table with the following columns: is_explicit_bool, release_quarter, is_weekend_release, log_stream_count, upbeat_score, artist_track_count, and Artist - Song Title. The table contains 21 rows of data.

	is_explicit_bool	release_quarter	is_weekend_release	log_stream_count	upbeat_score	artist_track_count	Artist - Song Title
1	FALSE	Q4	FALSE	10.7492	0.6632	3232	Ethereal-Storm
2	FALSE	Q4	TRUE	8.5895	0.7615	20	The Wolfs-Silent
3	FALSE	Q4	TRUE	12.2935	0.7828	199	LolaOre-Shattered Lines
4	FALSE	Q2	FALSE	8.5658	0.6778	693	Clay Black-Breaking Roses
5	FALSE	Q1	FALSE	10.2191	0.5494	12111	Knot Hart-Lines
6	FALSE	Q2	FALSE	9.8292	0.4586	12111	Knot Hart-Wild Doors
7	FALSE	Q3	FALSE	9.6072	0.5475	12111	Knot Hart-Skies
8	TRUE	Q4	FALSE	6.9745	0.2482	53	Blaze Project-Wild
9	TRUE	Q2	FALSE	8.3301	0.5249	693	Clay Black-Neon Oceans
10	FALSE	Q2	FALSE	12.1223	0.6488	242	Phantasm-Eternal Angels
11	TRUE	Q3	TRUE	9.1052	0.5284	12111	Knot Hart-Paradise
12	FALSE	Q3	FALSE	11.3219	0.6403	13	Zara Marsh-Edges
13	FALSE	Q4	TRUE	7.5005	0.6981	71	The Brookss-Circles
14	FALSE	Q1	TRUE	9.1957	0.6331	12111	Knot Hart-Burning Angels
15	FALSE	Q4	FALSE	6.7867	0.4417	12111	Knot Hart-The Roads
16	FALSE	Q2	TRUE	7.4366	0.5689	12111	Knot Hart-Velvet Memories
17	FALSE	Q4	FALSE	12.169	0.6151	5271	IsleEye-Gentle
18	FALSE	Q2	FALSE	9.4172	0.5639	1111	ElaraRain-Liquid Streets
19	FALSE	Q1	FALSE	11.7608	0.7445	2268	Sterling Zen-Broken Ghosts
20	FALSE	Q3	FALSE	10.5269	0.716	5271	IsleEye-Dancing Fire
21	TRUE	Q3	TRUE	11.5081	0.7305	635	Crest Jay-Dancing Roses

At the bottom left, it says "34 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows". At the bottom right, it says "PREVIEW DOWNLOADED AT 21:49".

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [2]

User_Reviews Zomato_Restaurants

Table.ReorderColumns(#"Removed Columns",{"Review_ID", "Restaurant_ID", "User_Name", "Review_Text", "Custom.1", "Rating"})

Restaurant_ID	User_Name	Review_Text	Custom.1	Rating	
1	101	Aarav Sharma	Great authentic North Indian food!	Average	4.5
2	101	Priya Singh	Good ambience but slow service.	Average	4
3	101	Rohan Mehta	Loved the butter chicken!	Top Rated	5
4	102	Ananya Roy	Pasta was decent but expensive.	Average	3.5
5	102	Vikram Verma	Nice rooftop seating.	Average	4
6	103	Siddharth Patel	Best dim sums in town!	Average	4.5
7	103	Neha Gupta	Quick delivery and fresh food.	Average	4
8	104	Kiran Joshi	Average food quality.	Average	3
9	104	Amit Kumar	Generous portion sizes.	Average	4
10	105	Sneha Nair	Delicious thin-crust pizza!	Top Rated	5

Query Settings

PROPERTIES

Name

User_Reviews

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Added Conditional Column

Added Conditional Column1

Removed Columns

Reordered Columns

6 COLUMNS, 10 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON 08 SEPTEMBER 2026

Type here to search

2:16 ENG IN 23-08 24-09-2026

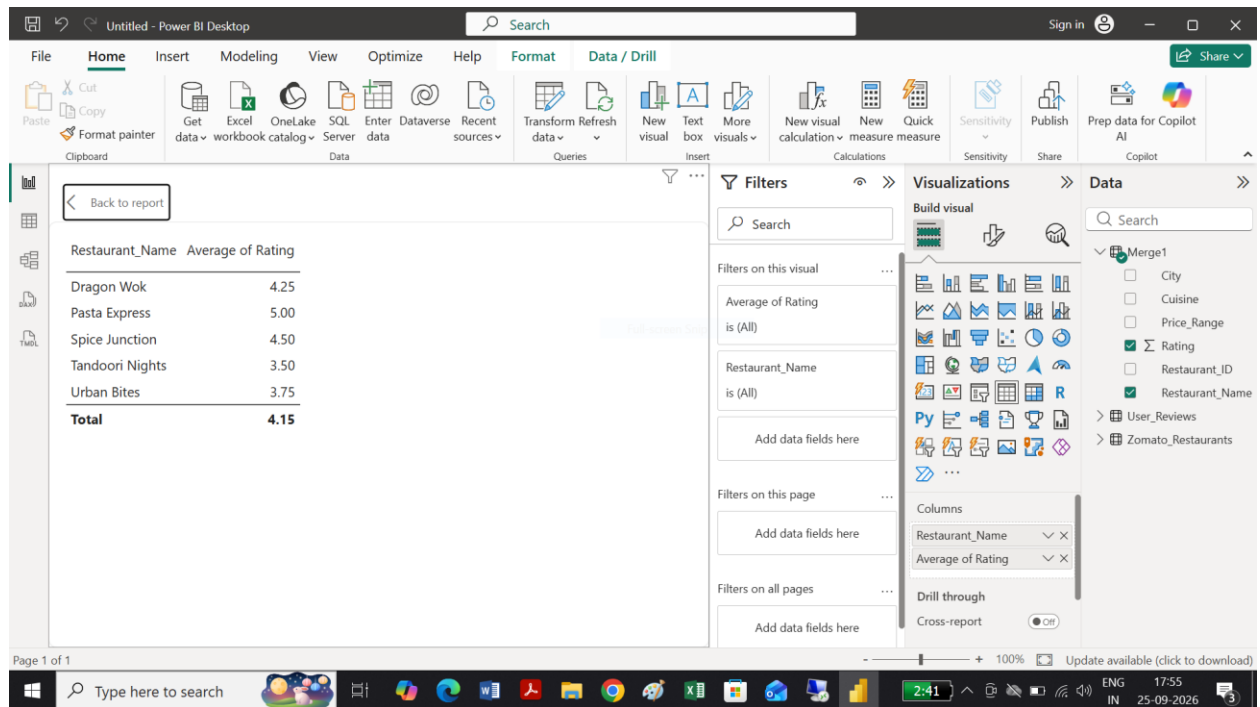
Task 5:



Session 6- Task
5.pbix

Session 7:

Task 1:



Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [3] Zomato_Restaurants User_Reviews Merge1

This preview may be up to 17 days old. Refresh

fx = Table.ExpandTableColumn(Source, "User_Reviews", {"Rating"}, {"Rating"})

	Restaurant_Name	City	Cuisine	Price_Range	Rating
1	Spice Junction	New Delhi	North Indian	Medium	4.5
2	Spice Junction	New Delhi	North Indian	Medium	4
3	Spice Junction	New Delhi	North Indian	Medium	5
4	Urban Bites	Mumbai	Italian	High	3.5
5	Urban Bites	Mumbai	Italian	High	4
6	Dragon Wok	Bengaluru	Chinese	Low	4.5
7	Dragon Wok	Bengaluru	Chinese	Low	4
8	Tandoori Nights	New Delhi	North Indian	Medium	3
9	Tandoori Nights	New Delhi	North Indian	Medium	4
10	Pasta Express	Pune	Italian	Low	5

6 COLUMNS, 10 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON 08 SEPTEMBER 2026

2:38 ENG IN 17:57 25-09-2026

Query Settings

PROPERTIES

Name

Merge1

All Properties

APPLIED STEPS

Source

Expanded User_Reviews

Task 2:



Session 7- Task
2.pbix

Untitled - Power BI Desktop

File Home Help Table tools

Name: User-Summary

Manage relationships | New Quick measure measure column | New table | Mark as date table | Calendars

Structure

user_id	Total Orders	Total Amount Spent
USER_101	4	1500
USER_103	2	560
USER_105	1	400
USER_102	3	520
USER_104	2	1450

Data

Search

- > All Orders
- > Last_Month_Orders
- > This_Month_Orders
- > User-Summary

Table: User-Summary (5 rows)

Update available (click to download)

Task 3:



Session 7- TASK
3.pbix

Session 7- TASK 3 • Last saved: Today at 8:09 PM

File Home Help Table tools

Name: data

Manage relationships | New Quick measure measure column | New table | Mark as date table | Calendars

Structure

Source.Name	Player	Balls Faced	Runs Scored
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Virat Kohli	12	18
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Rohit Sharma	15	22
IPL_Player_Balls_Faced_Runs_Scored.xlsx	KL Rahul	20	31
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Shubman Gill	10	14
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Suryakumar Yadav	18	28
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Rishabh Pant	16	25
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Jos Buttler	22	36
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Hardik Pandya	14	20
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Ruturaj Gaikwad	19	29
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Sanju Samson	17	24
IPL_Player_Balls_Faced_Runs_Scored.xlsx	David Warner	13	17
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Shreyas Iyer	21	33
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Ravindra Jadeja	11	15
IPL_Player_Balls_Faced_Runs_Scored.xlsx	MS Dhoni	9	12
IPL_Player_Balls_Faced_Runs_Scored.xlsx	Ishan Kishan	16	23

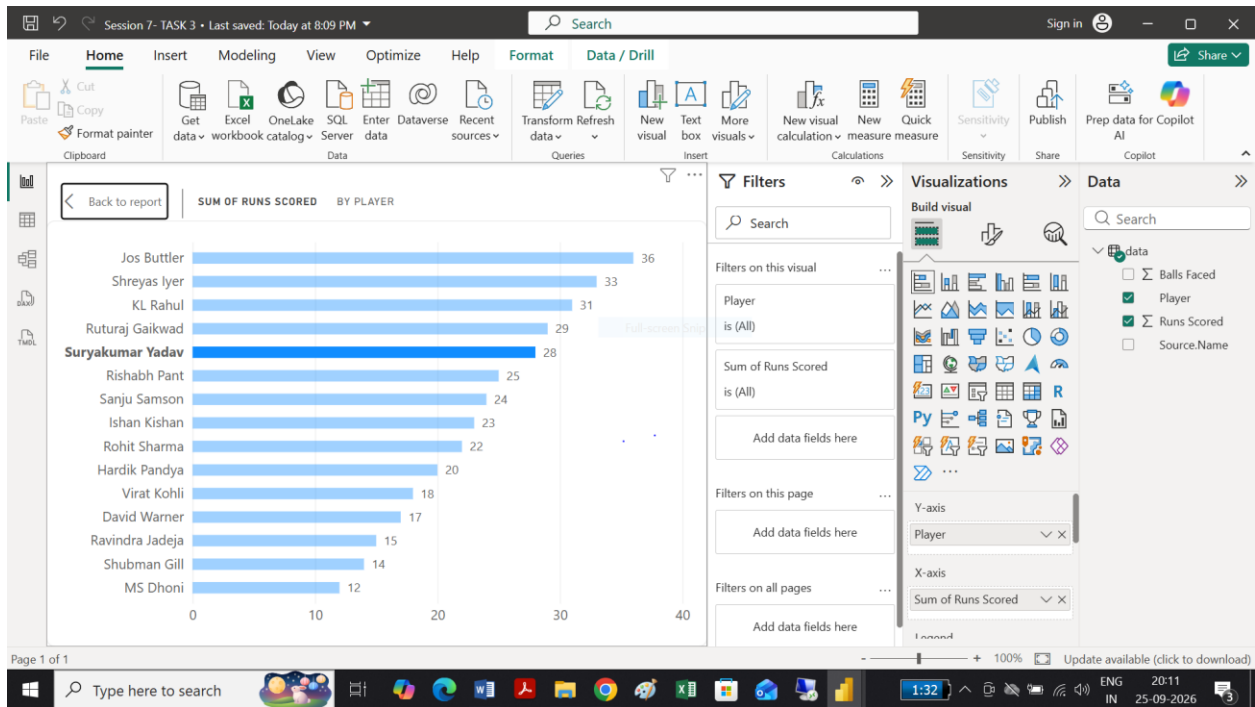
Data

Search

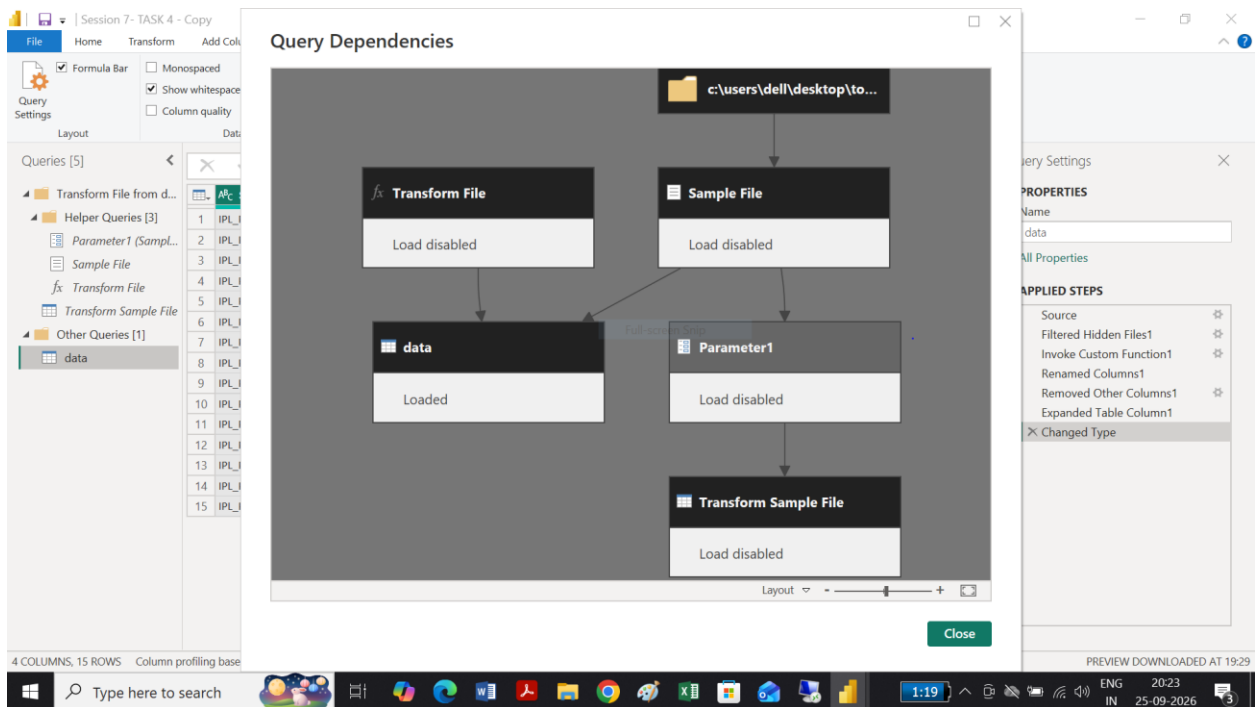
- > data

Table: data (15 rows)

Update available (click to download)



Task 4:



Session 8:

Task 1:



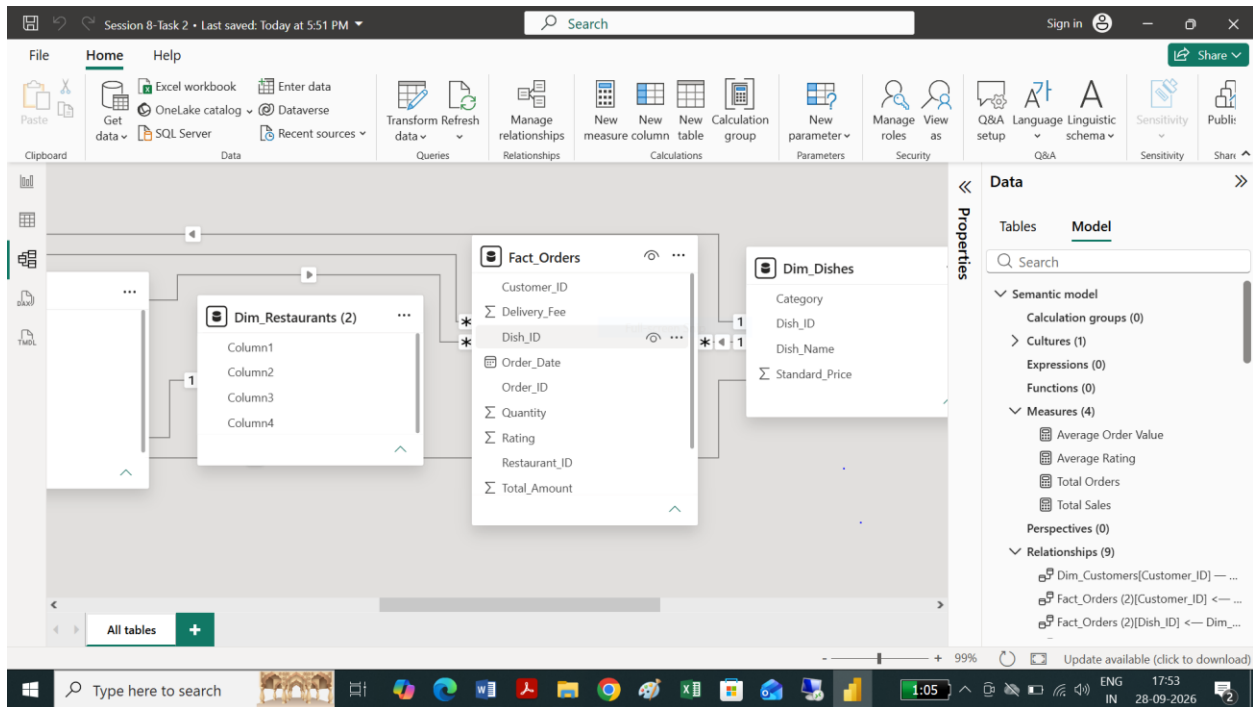
Session 8. Task 1.pbix

The screenshot displays the Microsoft Power BI Desktop interface. The main workspace shows a data model with two tables: 'Restaurants' and 'ReviewsS'. The 'Restaurants' table has columns for 'Cuisine', 'Name', and 'RestaurantID'. The 'ReviewsS' table has columns for 'Comment', 'Rating', 'RestaurantID', and 'ReviewID'. A relationship is established between the 'RestaurantID' columns of both tables, with a cardinality of 'Many to one (*:1)'. The 'Make this relationship active' checkbox is checked. The 'Cross-filter direction' is set to 'Single'. The 'Apply security filter in both directions' checkbox is unchecked. The 'Properties' pane on the right shows the relationship details. The 'Data' pane on the right shows the semantic model structure, including 'Tables (4)' and 'Relationships (1)'. The bottom status bar shows the time as 0:54 and the date as 25-09-2026.

Task 2:



Session 8-Task 2.pbix

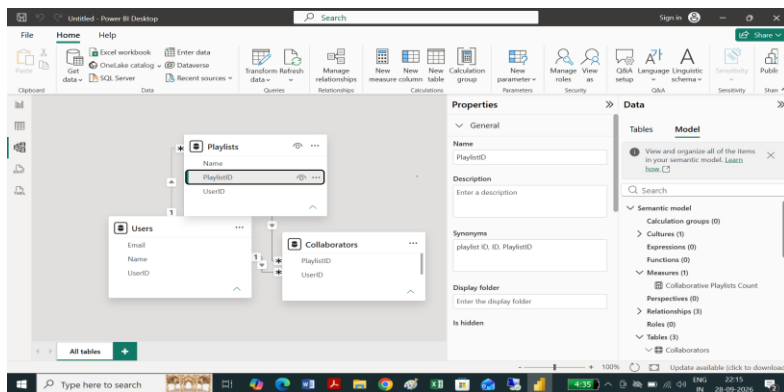


Task 3:



Session 8- Task
3.pbix

Task 4:



Task 5:



Session 8- Task
5.pbix

Session 8- Task 5 • Last saved: Today at 5:03 PM

Search

File Home Insert Modeling View Optimize Help Format Data / Drill

Clipboard: Cut, Copy, Paste, Format painter

Data: Get data, Excel, OneLake, SQL Server, Enter data, Dataverse, Recent sources

Queries: Transform, Refresh, data

Insert: New visual, Text box, More visuals

Calculations: New visual calculation, New measure, Quick measure

Sensitivity: Sensitivity, Publish, Prep data for Copilot

Visualizations: Filters, Data

Back to report

OrderID	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010
DishName	2026	Total	2026	Total	2026	Total	2026	Total	2026	Total
Caesar Salad			1	1						
Garlic Bread								1	1	
Grilled Salmon								1	1	
Iced Tea						1	1			
Margherita Pizza	1	1		1	1				1	1
Tiramisu					1	1				
Total	1	1	1	1	1	1	1	1	1	1

Page 1 of 1

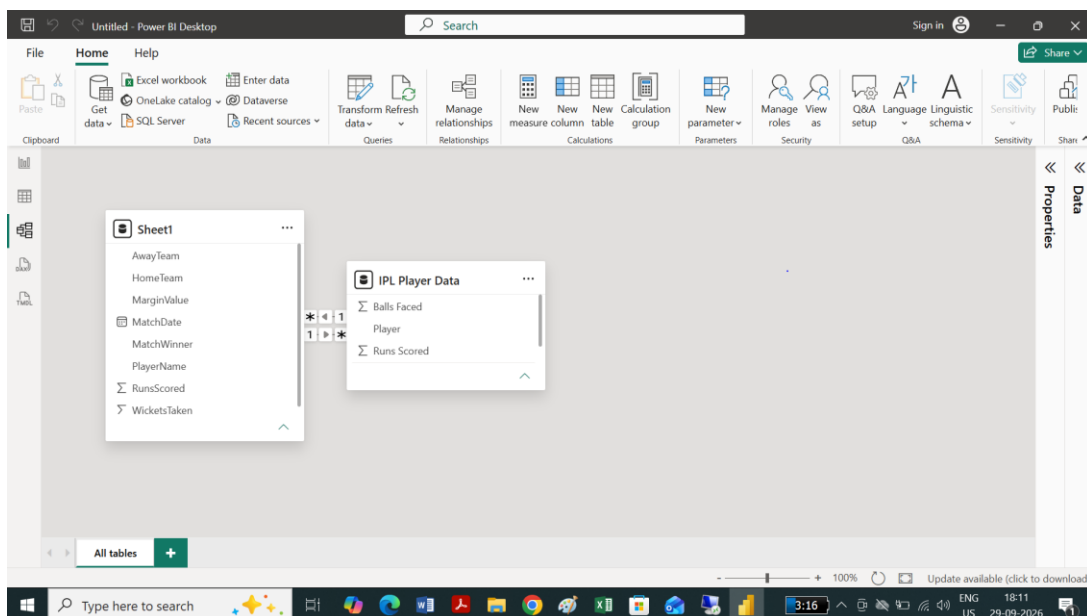
Type here to search

1:18

ENG US 17:04 29-09-2026

Session 9:

Task 1:



Explanation of Inactive Relationships in Sports Analytics

Why Inactive Relationships are Essential in Sports Analytics:

1. **Handling Role-Based / Multiple Foreign Keys (Role-Playing Dimensions):** A single cricket match inherently involves a team in multiple roles—as the **Home Team**, **Away Team**, or **Winning Team**. Power BI only allows **one active relationship path** between two tables at a time to prevent circular references and ambiguous filter contexts.

2. **Flexible & Dynamic Measure Calculations:** By keeping additional relationships (like `Teams[Team_ID] ↔ Matches[Away_Team_ID]`) **inactive**, you preserve the integrity of the data model while retaining full analytical capability. You can dynamically activate the secondary relationship inside specific DAX measures using the `USERELATIONSHIP()` function to compute metrics such as:
 - **Home vs. Away Performance Split:** Easily compare home wins vs. away wins without duplicating the Teams table.
 - **Head-to-Head & Match Analysis:** Filter performance metrics depending on whether the team was batting first, chasing, playing at home, or away.

Task 2:

Here is how this change affects your report when you select a specific match:

1. The Core Effect (Bi-Directional Filtering)

- **Under 'Single' Direction:** The relationship typically filters in one direction—usually from the dimension table to the fact table, or along the direction of the arrow (e.g., *Players* → *Matches* or vice versa). Selecting a match only filters tables downstream from *Matches*.
- **Under 'Both' Direction:** The filter context now flows in **both directions**. Selecting a match in your report filters the *Matches* table, which in turn filters the *Players* table, which then propagates down to filter any related tables like *Player Stats* (or performance metrics).

2. What You Observe in Your Report

When you select or slice by a specific match (e.g., *Match ID: 101*):

1. **Filtered Player List:** The *Players* table is automatically reduced to **only show the players who participated in that specific match**. Unrelated players are filtered out.
2. **Contextual Player Stats:** Visuals, tables, or cards displaying player statistics (e.g., goals scored, total passes, minutes played, or points) will automatically calculate metrics **strictly for that specific match** for the filtered subset of players.
3. **Slicer Interactivity:** If you have a player dropdown/slicer on the report page alongside a match slicer, selecting a match will immediately shrink the player slicer options so it only displays players relevant to that match.

💡 Things to Keep in Mind

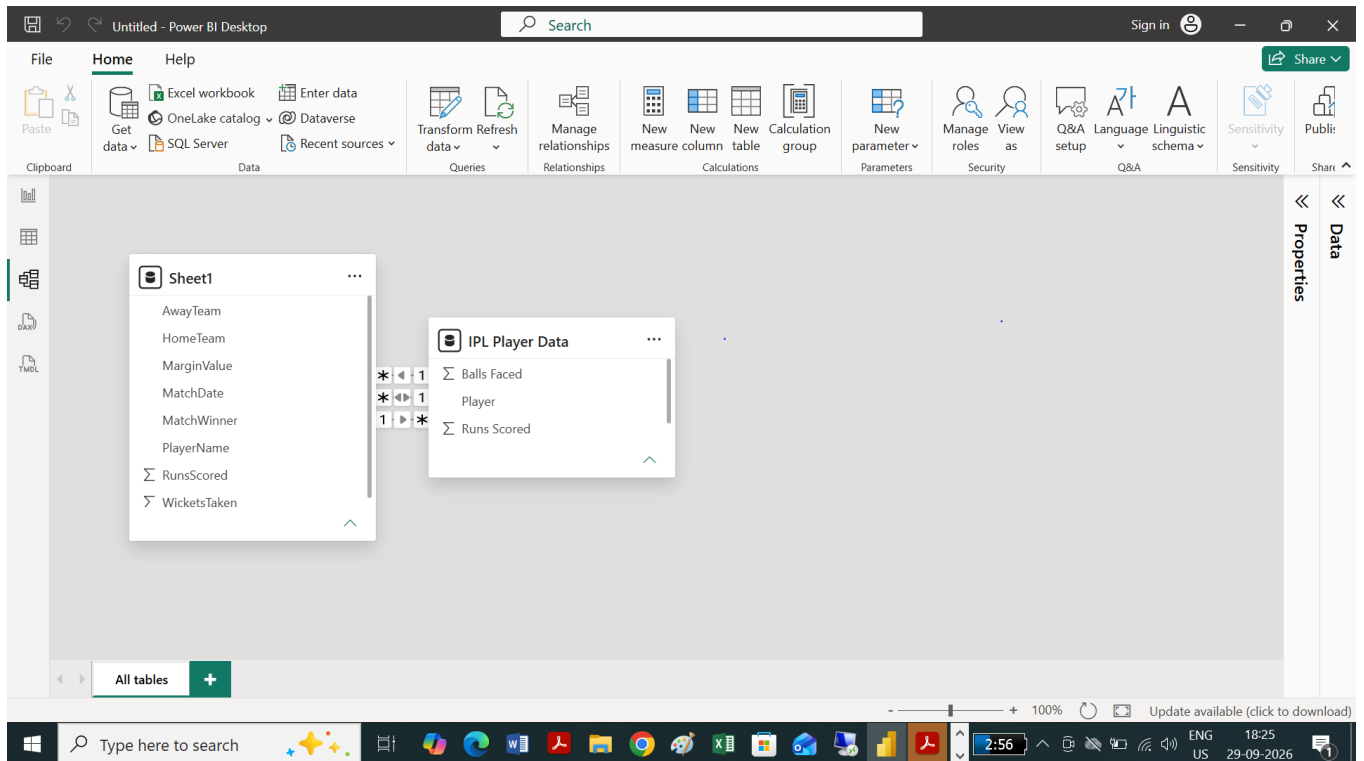
- **Ambiguity and Performance:** While setting cross-filter direction to "Both" provides intuitive filtering across related entities, using bi-directional filtering across multiple tables can lead to ambiguous model paths or slower report performance in large models.
- **Alternative Approach:** In production models, DAX functions like `CROSSFILTER()` inside a measure or using bridge/fact-table filters are often preferred over setting the model-level relationship to "Both."



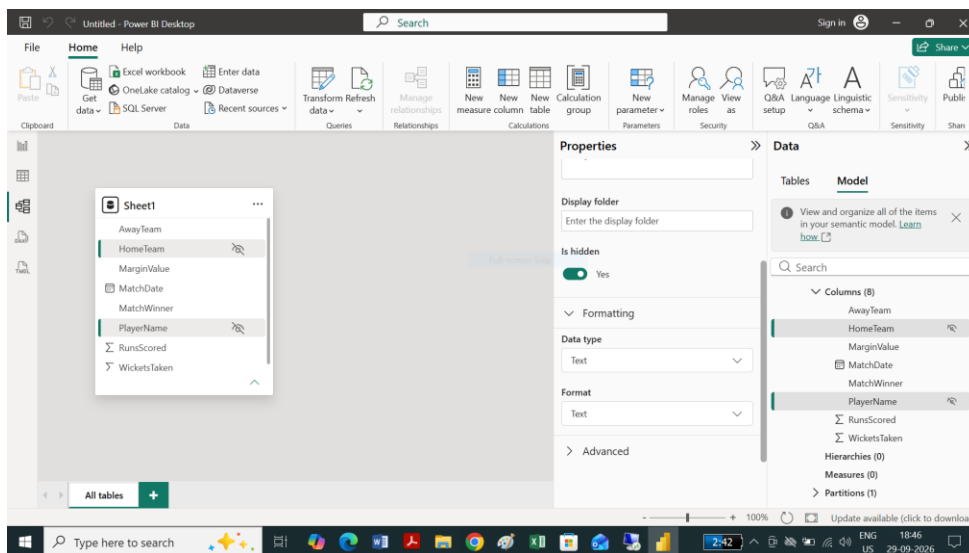
session 9- task 2.pbix

The screenshot displays the Microsoft Power BI Desktop application. The main window shows a table titled "IPL Player Data" with 15 rows and 3 columns: "Player", "Balls Faced", and "Runs Scored". The table lists players such as Virat Kohli, Rohit Sharma, KL Rahul, Shubman Gill, Suryakumar Yadav, Rishabh Pant, Jos Buttler, Hardik Pandya, Ruturaj Gaikwad, Sanju Samson, David Warner, Shreyas Iyer, Ravindra Jadeja, MS Dhoni, and Ishan Kishan. The interface includes a ribbon with tabs for File, Home, Help, Table tools, and Measure tools. The Measure tools tab is active, showing options for Name, Home table, Format, and Data category. The Data pane on the right shows the hierarchy of the data model, including the "IPL Player Data" table and its columns. The status bar at the bottom indicates the table has 15 rows and 3 columns, and provides a link to download the latest update.

Player	Balls Faced	Runs Scored
Virat Kohli	12	18
Rohit Sharma	15	22
KL Rahul	20	31
Shubman Gill	10	14
Suryakumar Yadav	18	28
Rishabh Pant	16	25
Jos Buttler	22	36
Hardik Pandya	14	20
Ruturaj Gaikwad	19	29
Sanju Samson	17	24
David Warner	13	17
Shreyas Iyer	21	33
Ravindra Jadeja	11	15
MS Dhoni	9	12
Ishan Kishan	16	23



Task 3: Home team and Player name team is hidden



Key Notes & Security Considerations

- **Visual Verification:** Hidden fields will appear grayed out in Power BI Desktop's Data pane to remind you they are hidden, but they will be completely invisible to report viewers once published to the Power BI Service.
- **Calculations:** Measures and DAX expressions will still be able to reference `PlayerEmail` or `PlayerPhone` in the background even when they are hidden from the user interface.
- **Important Security Note:** Hiding a column in Report View hides it from the UI, but it **does not secure the data**. Users with permission to download the .pbix file or connect via Analyze in Excel can still access hidden columns. If you need true security to restrict access to PII (Personally Identifiable Information), implement **Row-Level Security (RLS)** or remove the columns in Power Query before loading the data model.

Task 4:



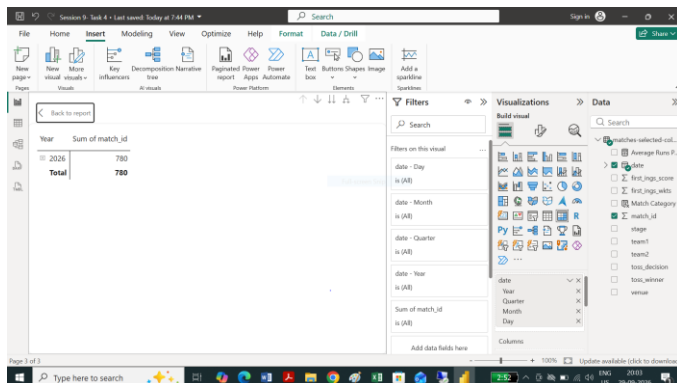
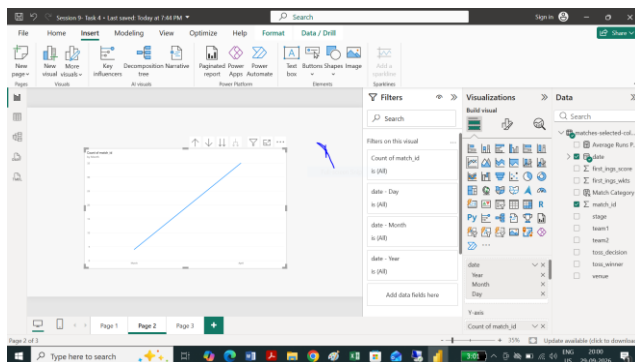
Session 9- Task
4.pbix

Key Behavioral Differences: Calculated Columns vs. Measures

Feature	Calculated Column (Match Category)	Measure (Average Runs Per Match)
Evaluation Context	Row Context: Evaluated line-by-line during data refresh.	Filter Context: Evaluated dynamically based on active visual filters, slicers, or user interactions.
Storage & RAM	Stored directly in the Data Model, consuming disk space and RAM memory.	Not stored in memory; calculated on-the-fly when added to a visual or report.

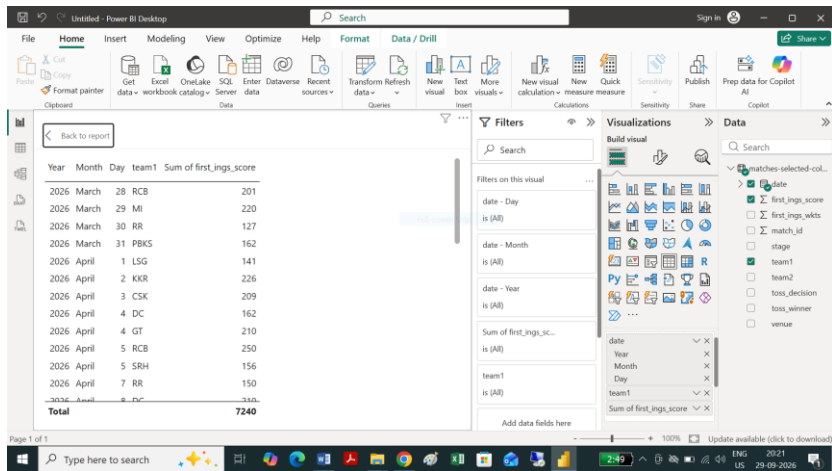
Feature	Calculated Column (Match Category)	Measure (Average Runs Per Match)
Usage in Visuals	Ideal for slice-and-dice operations (Slicers, Row/Column Headers, Axis categories).	Ideal for numerical values, aggregations, KPIs, tooltips, and data cards.
Reactivity	Static; does not react dynamically to slicers or report interactions.	Dynamic; automatically updates value whenever filters or slicers change.

Task 5:

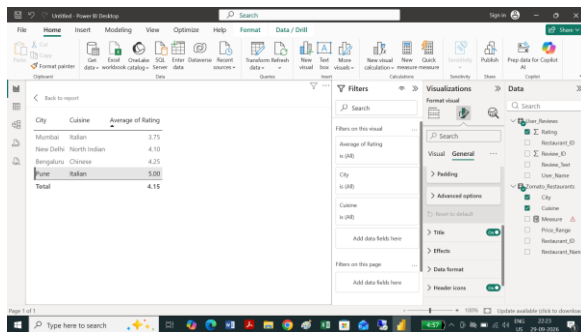


Session 10:

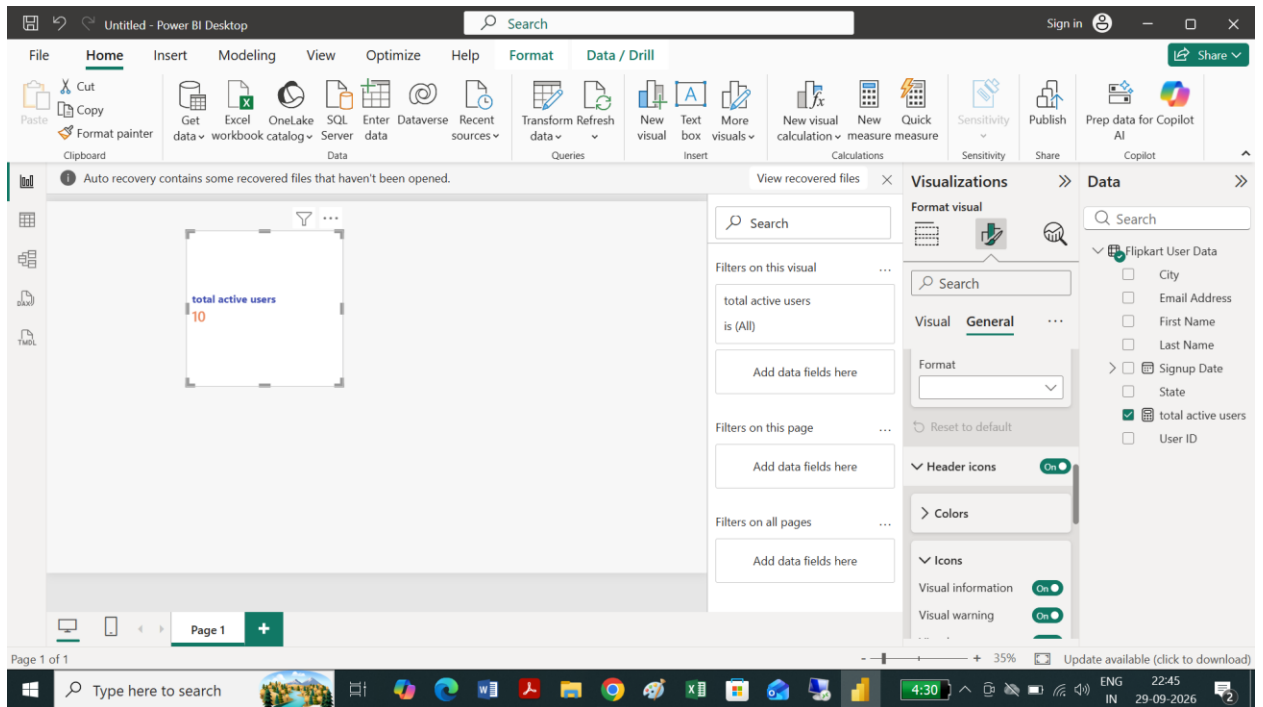
Task 1:



Task 2:

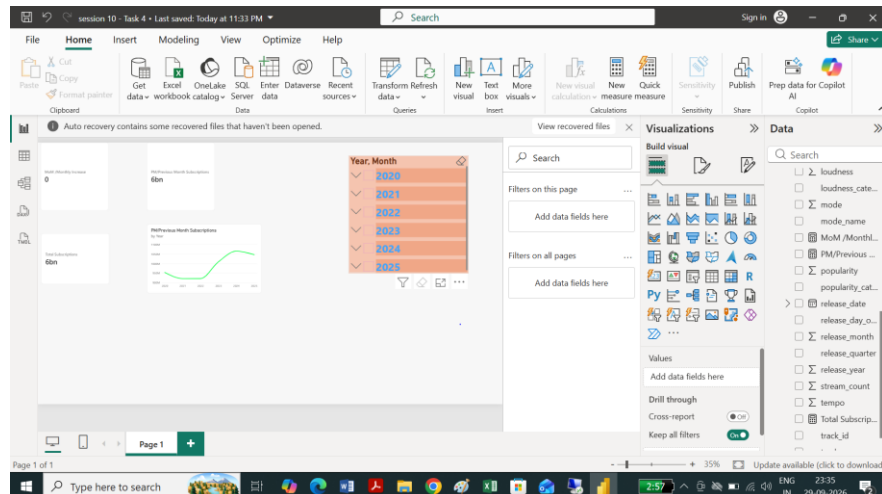


Task 3:



Task 4:

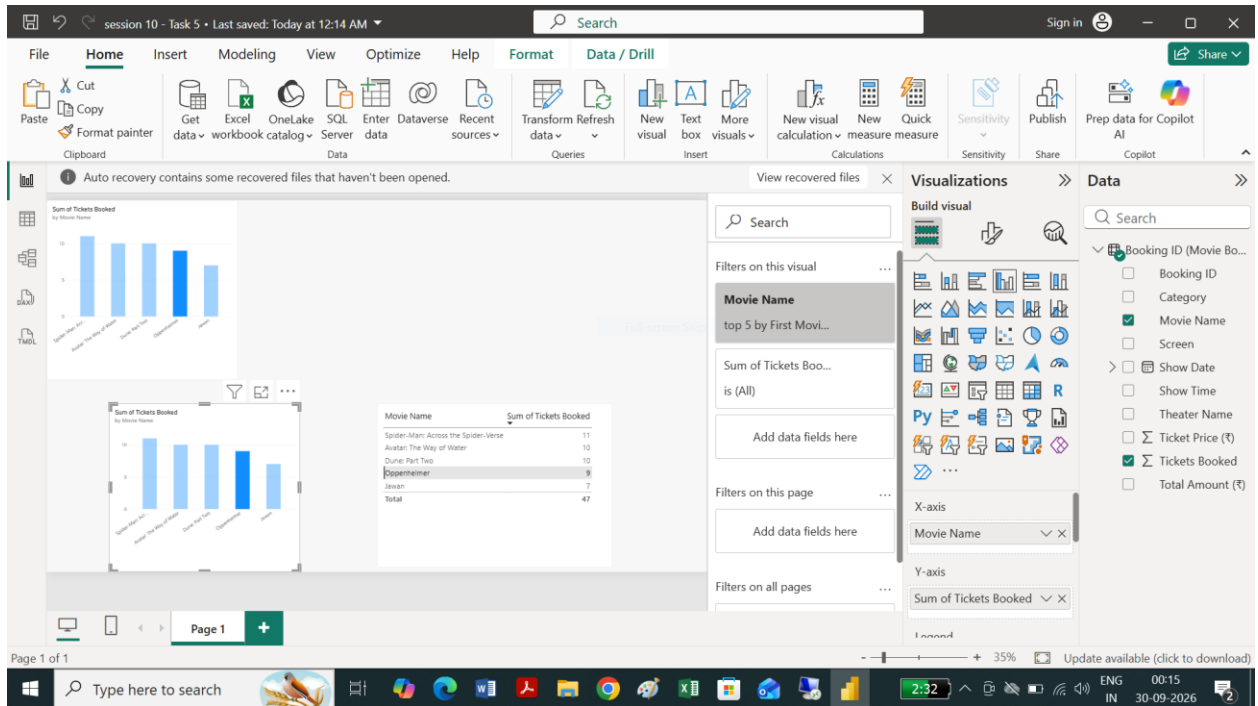

session 10 - Task
4.pbix



Task 5:



session 10 - Task
5.pbix



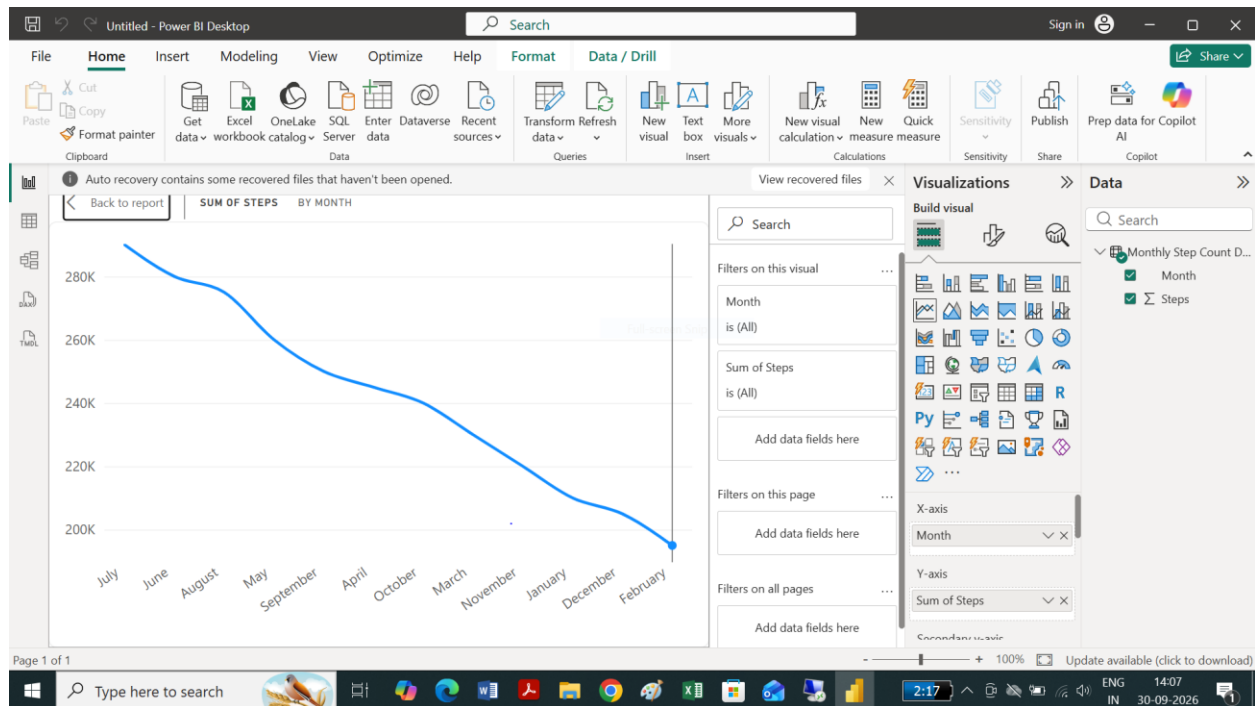
session 11



monthly_step_count_
data.xlsx

Task 1:

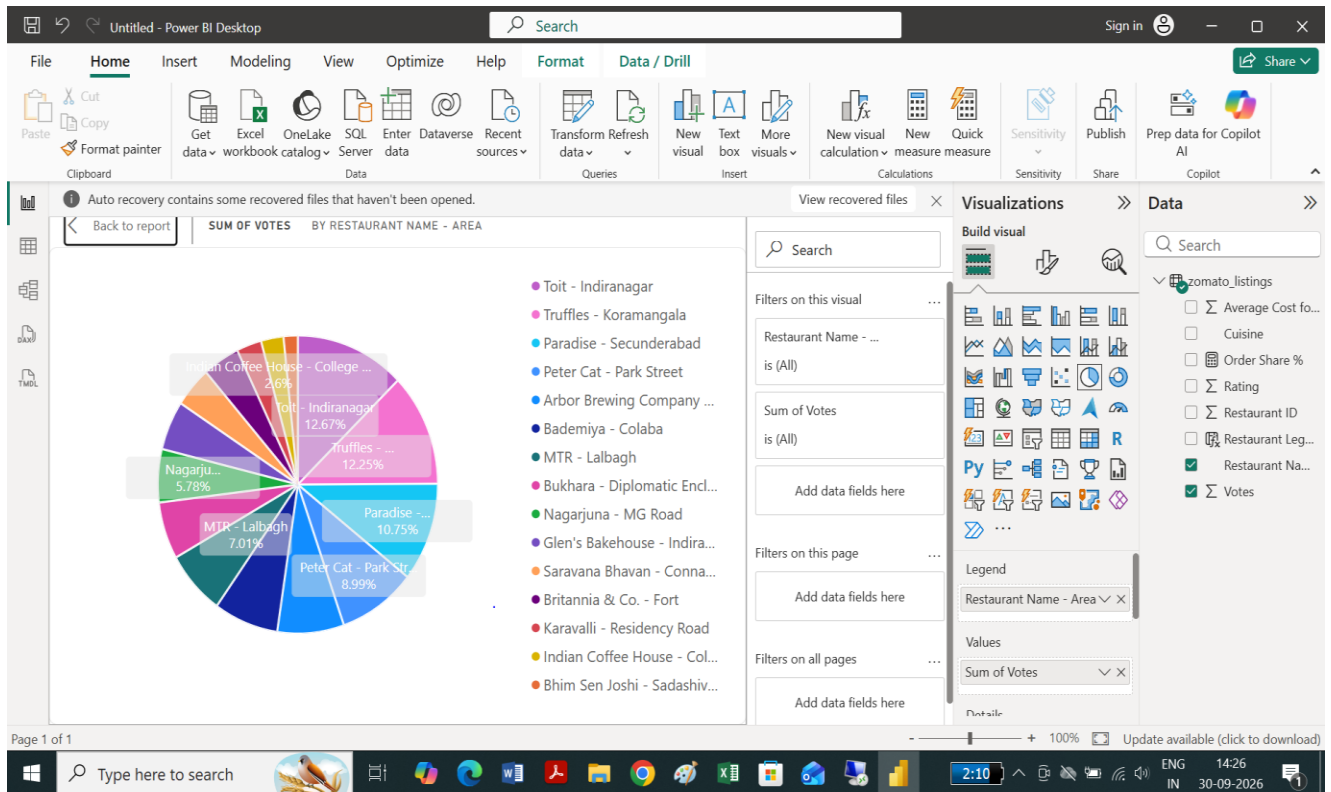
Below results seen from above data attached



Task 2:



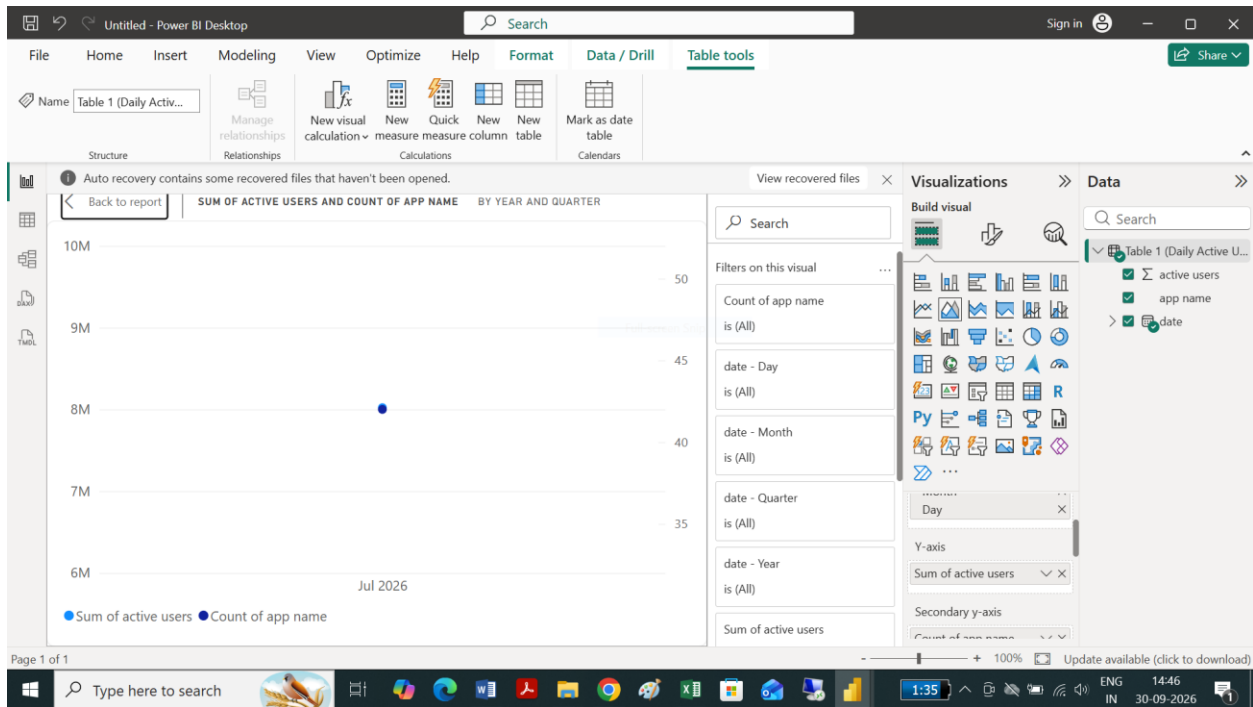
session 11- task
2.pbix



Task 3:



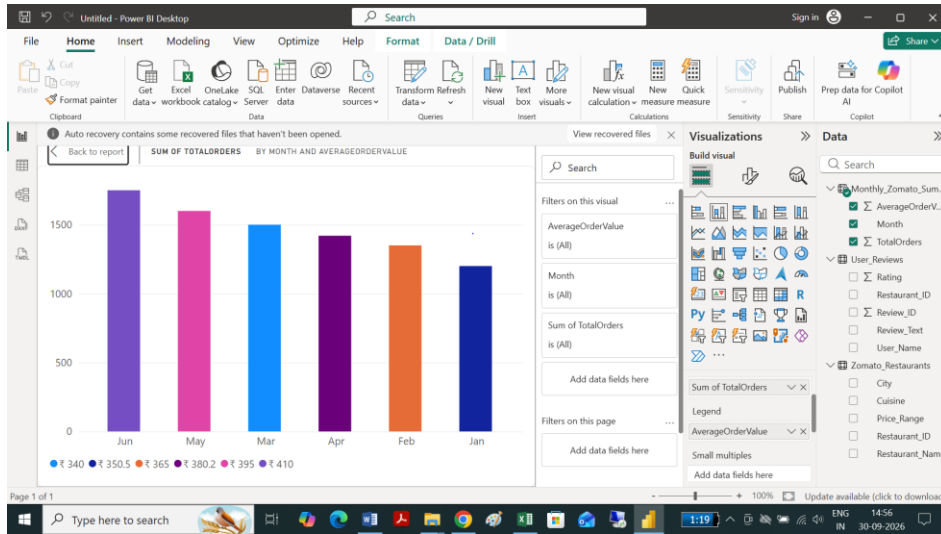
session 11- task
3.pbix



Task 4:



session 11- task
4.pbix

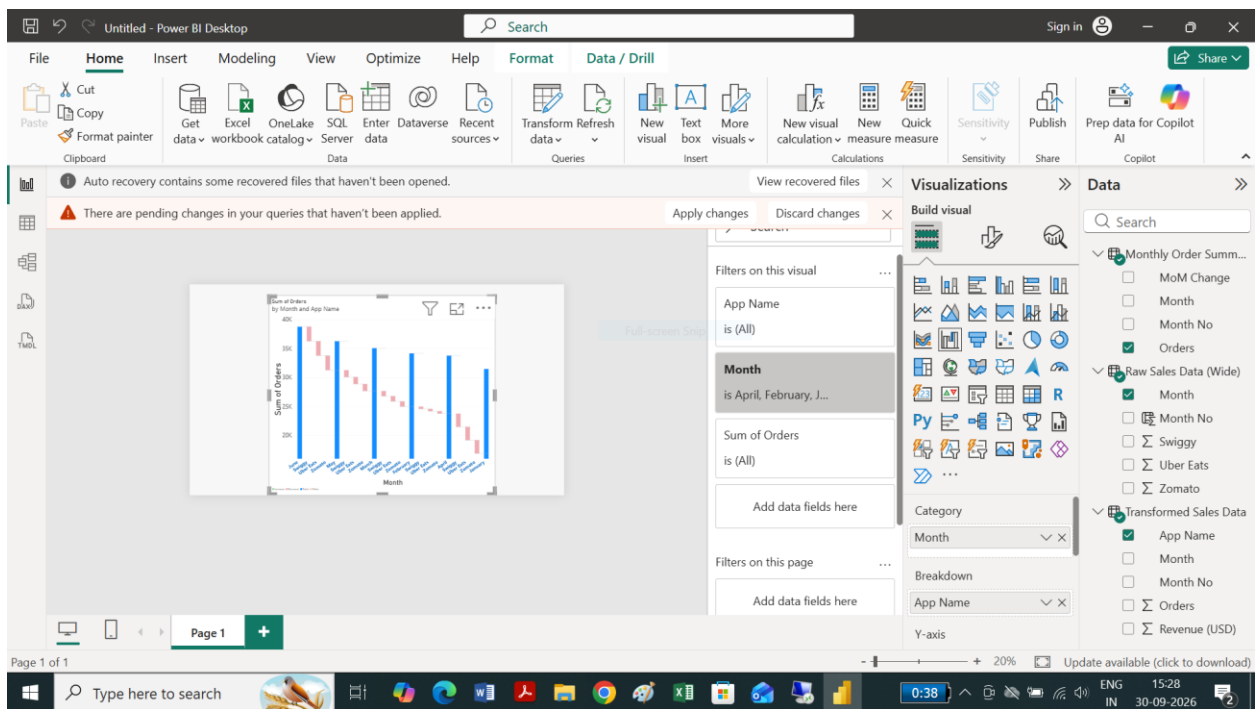


Session 12:

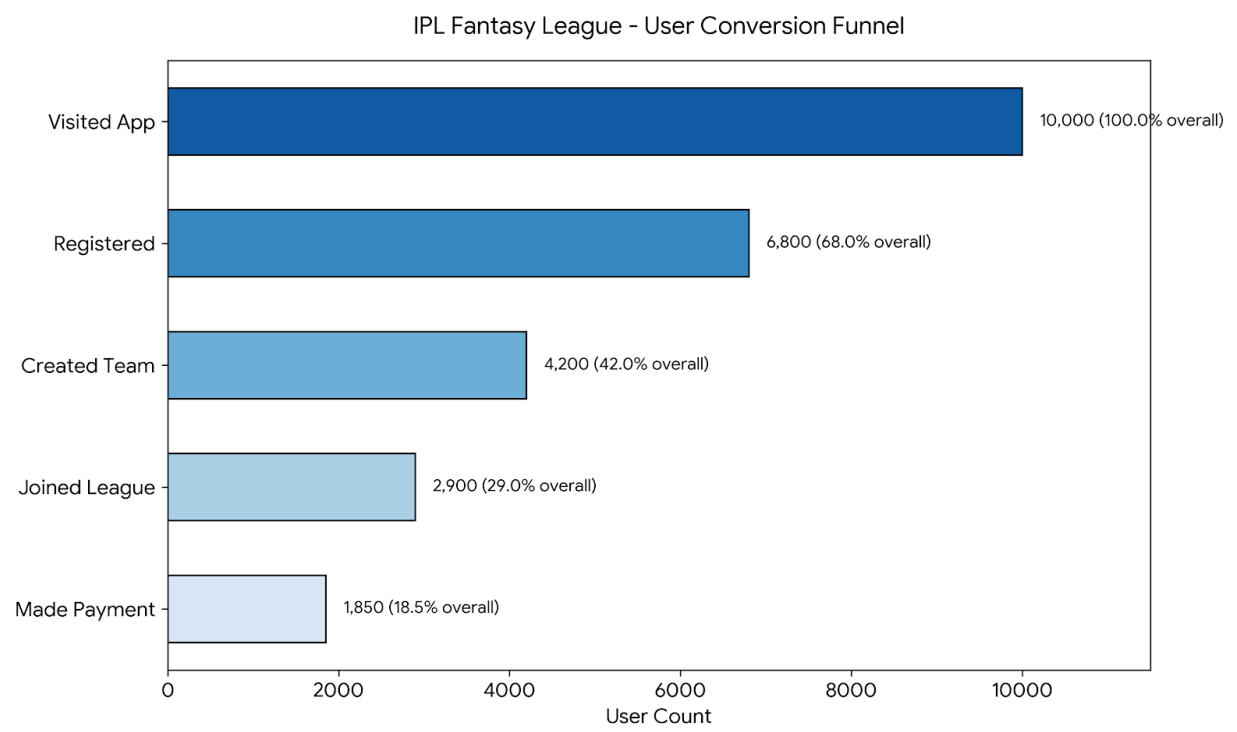
Task 1:



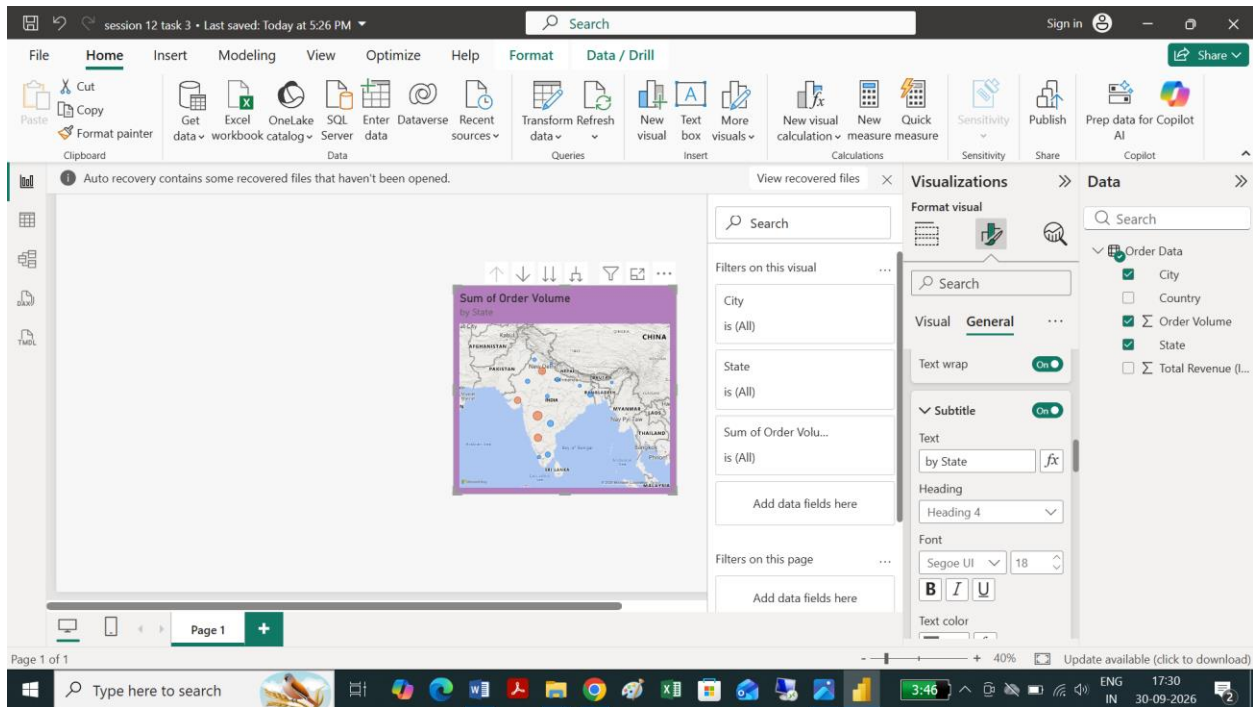
session 12- Task
1.pbix



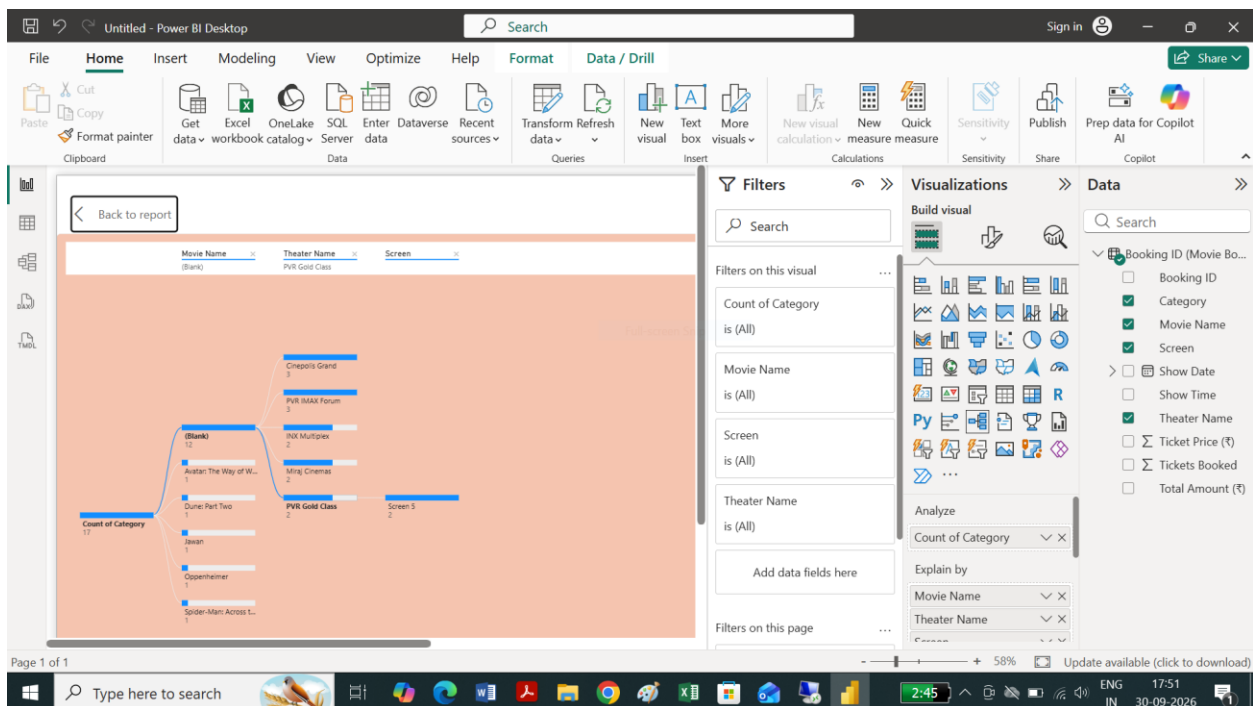
Task 2:



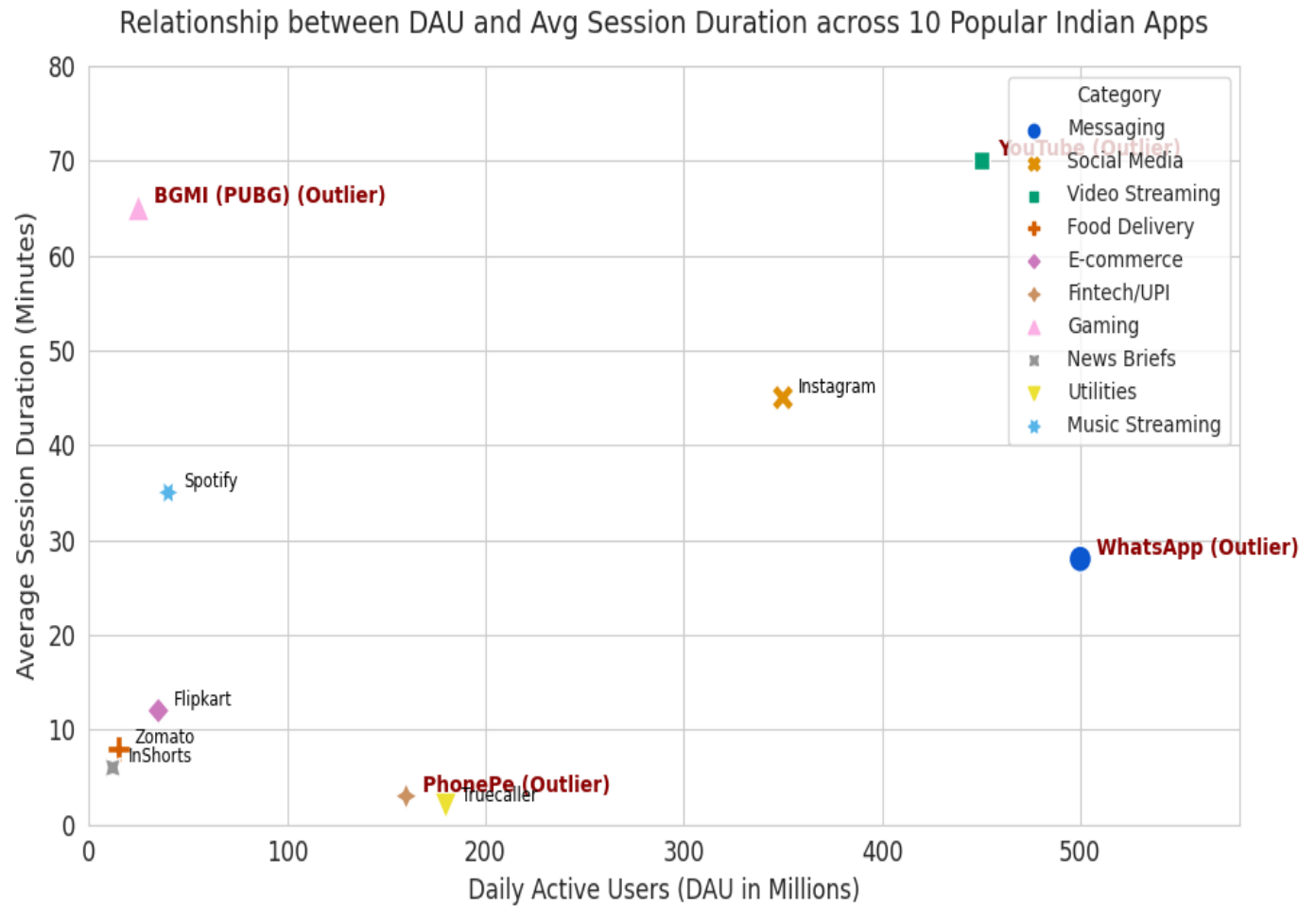
Task 3:



Task 4:

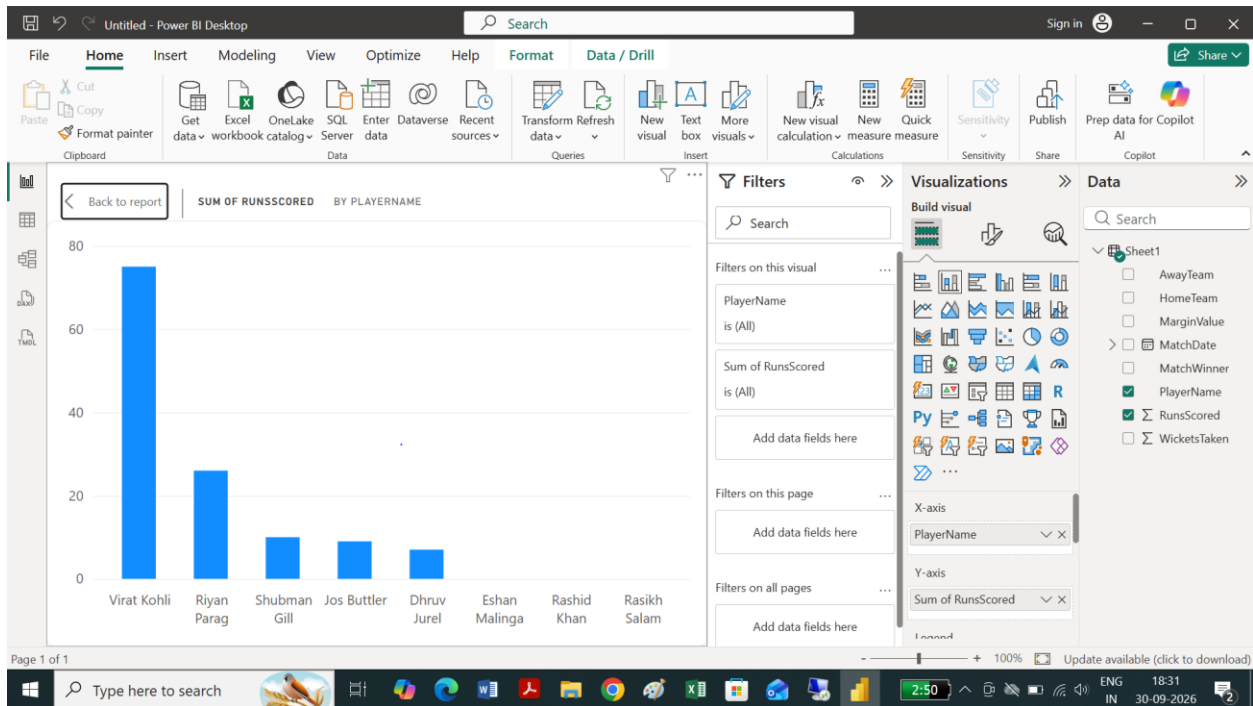


Task 5:

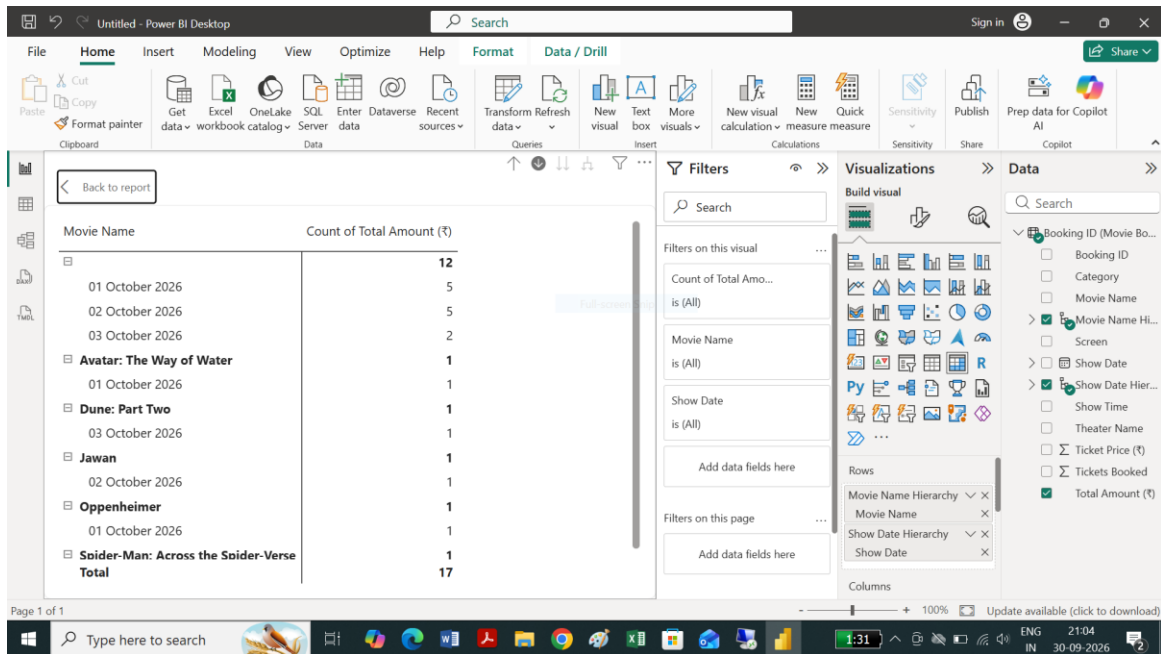


Session 13:

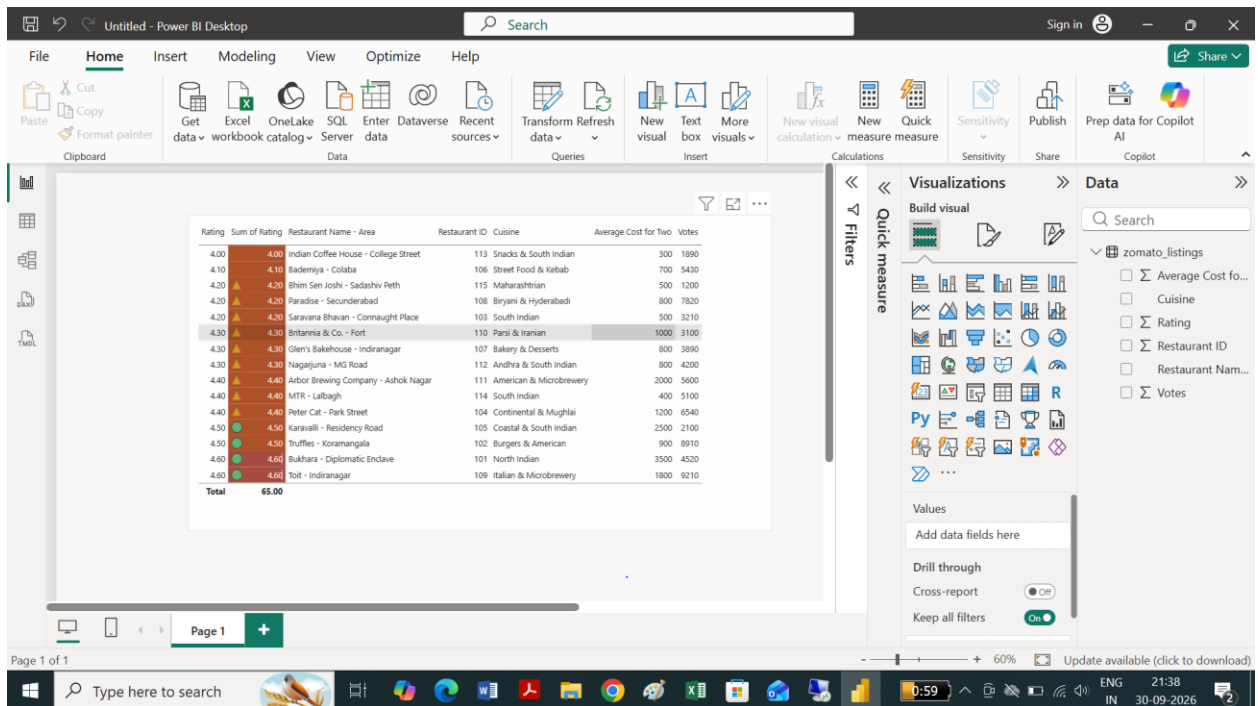
Task 1:



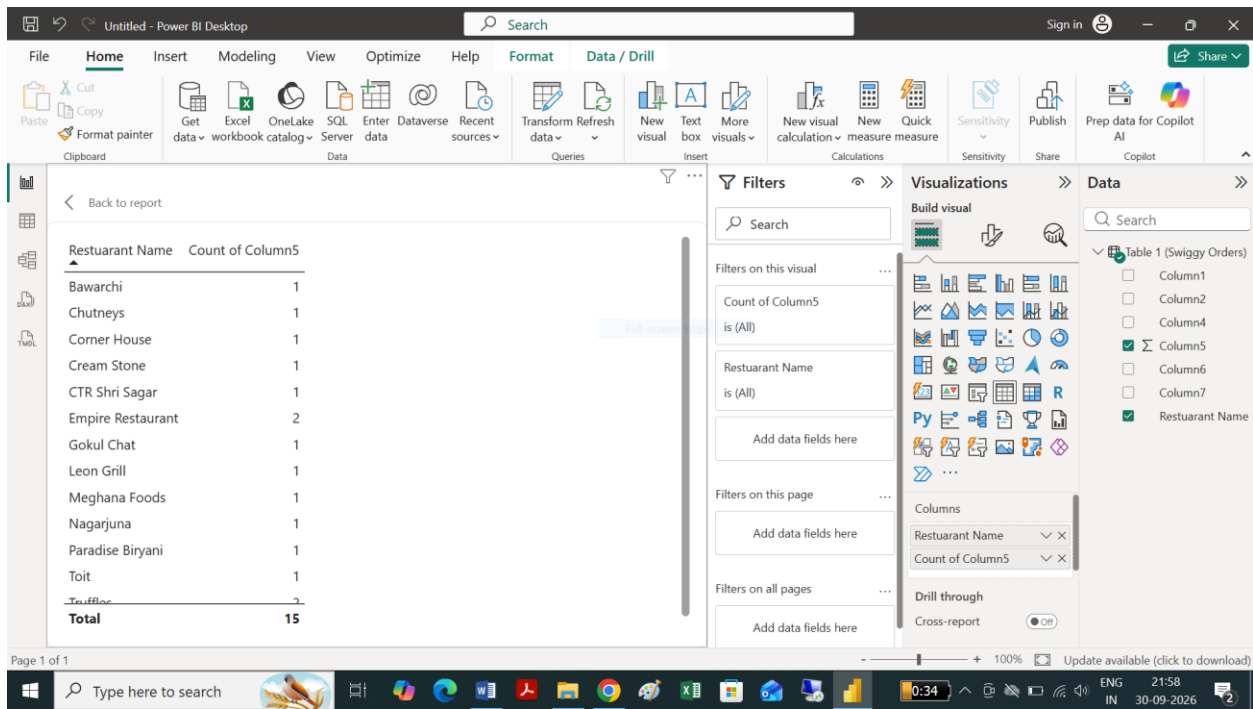
Task 2:



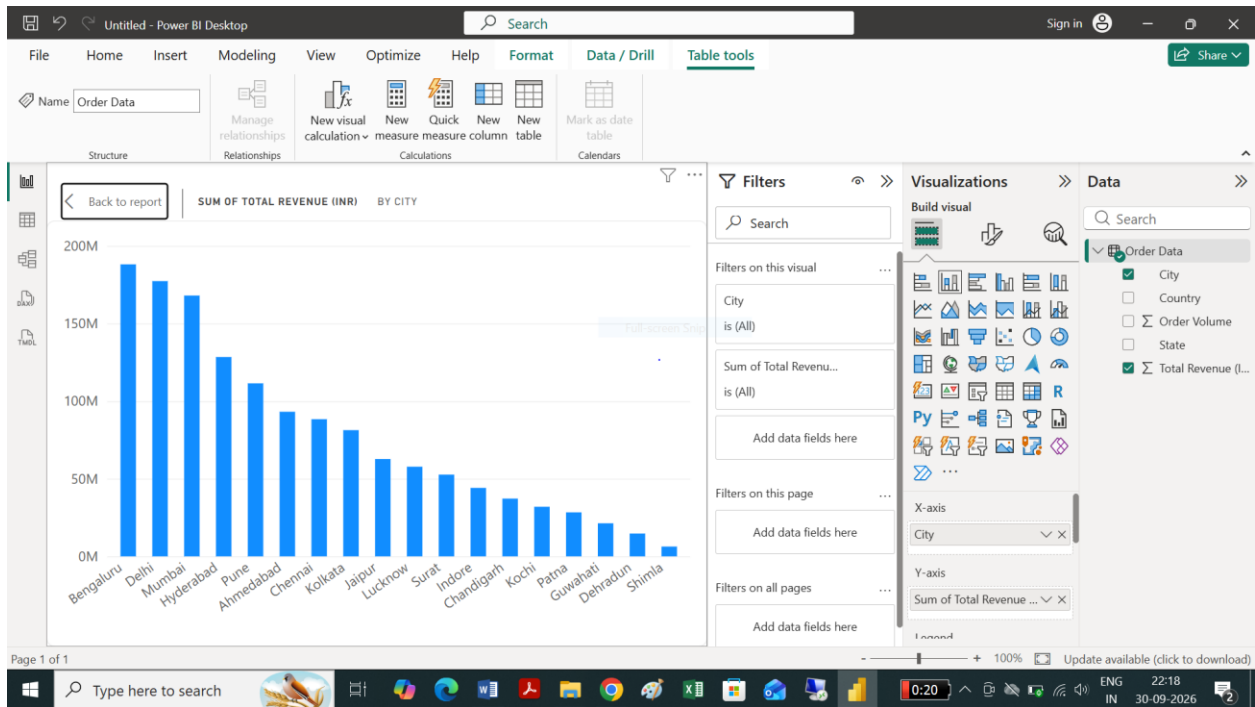
Task 3:



Task 4:

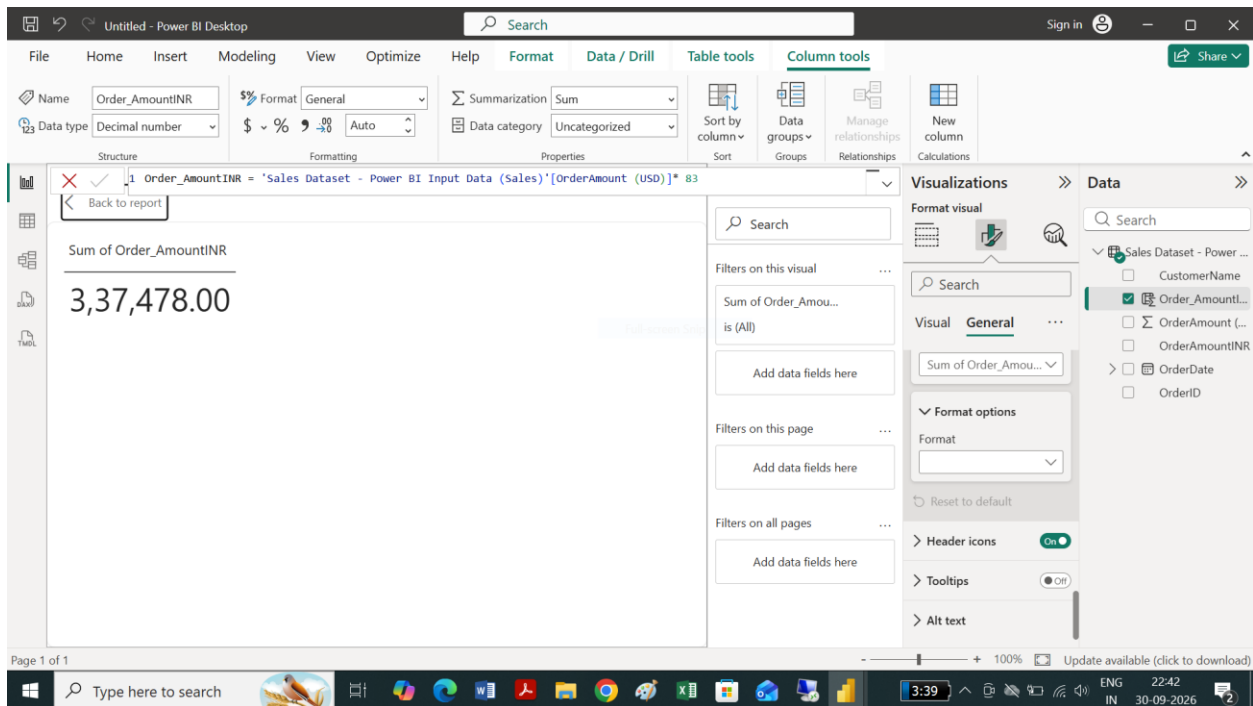


Task 5:



Session 14:

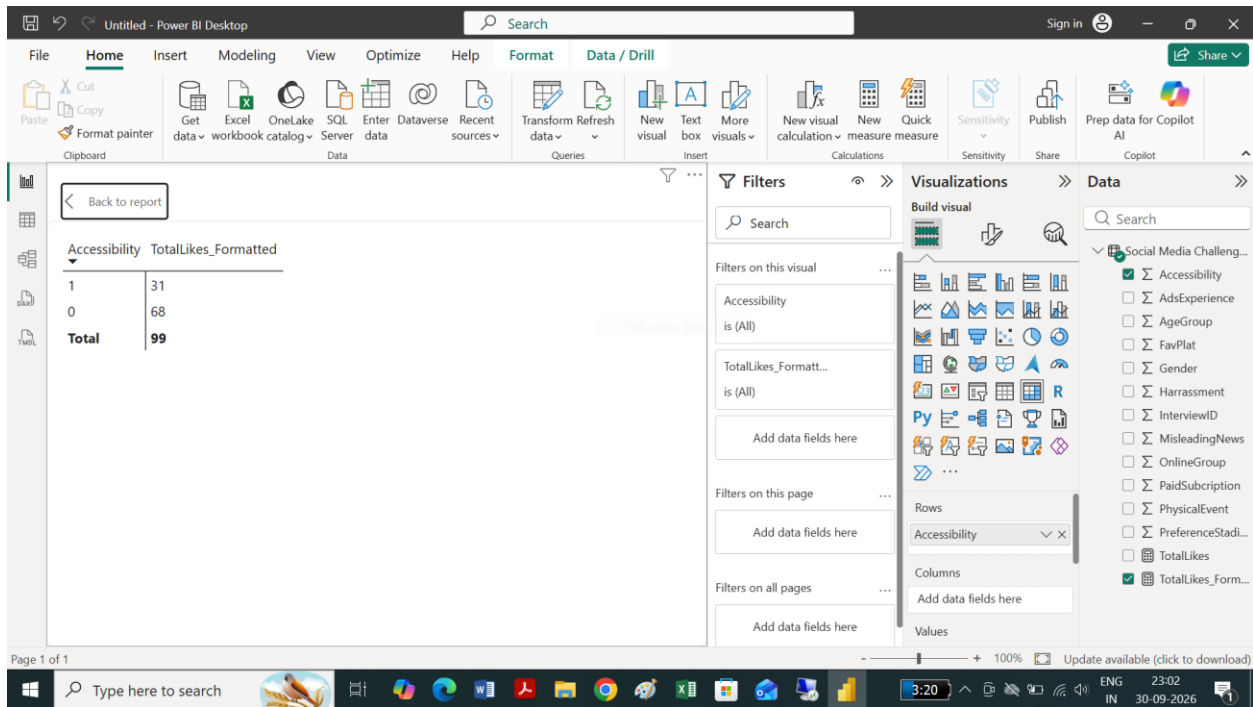
Task 1:



Task 2:

Measure: Totallikes_Formatted =

```
VAR _Likes = SUM('Social Media Challenges'[AdsExperience])
RETURN
    SWITCH(
        TRUE(),
        _Likes >= 1000000, FORMAT(_Likes / 1000000, "#,##0.0") & "M",
        _Likes >= 1000, FORMAT(_Likes / 1000, "#,##0.0") & "K",
        FORMAT(_Likes, "#,##0")
    )
```

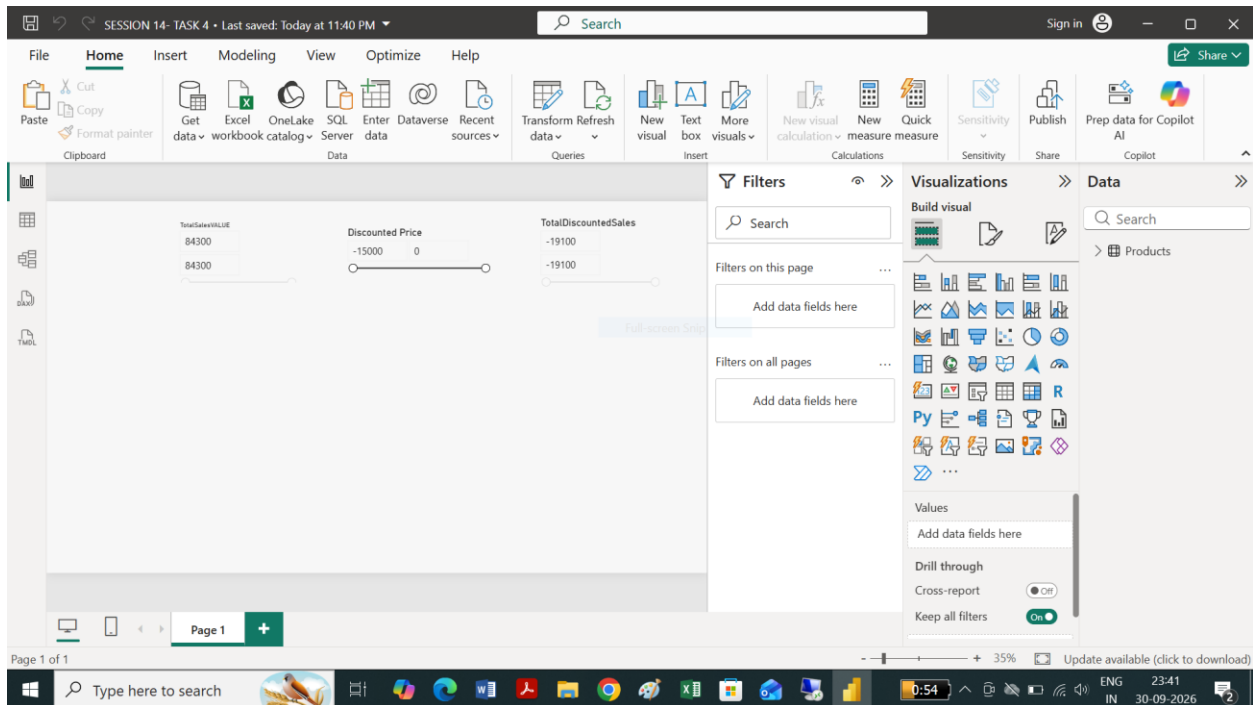



session 14 - task
2.pbix

Task 3:

Task 4:


SESSION 14- TASK
4.pbix



Task 5:

DAX Data Type Previous DAX Fields / Values Used Description Real-World App Field

Integer (Whole Number / `int64`) Quantity (e.g., 2), OrderID (e.g., 101) 64-bit integer numbers used for exact counts, identifiers, or quantities. **Zomato:** ItemQuantity in cart (e.g., 2 Butter Naans) or DeliveryRiderID. **Currency (Fixed Decimal / `currency`)** UnitPrice (15000), Discount (1000), DiscountedPrice (14000) Exact decimal numbers formatted with fixed precision (up to 4 decimal places) to prevent rounding errors in money/financial calculations. **Paytm:** WalletBalance, TransactionAmount, or CashbackAmount (e.g., ₹250.50). **Decimal / Double (Floating Point)** Measure results like TotalDiscountedSales when using dynamic formulas/ratios Floating-point decimal numbers used for rates, percentages, continuous calculations, and dynamic aggregates. **Zomato / Swiggy:** RestaurantRating (e.g., 4.3) or DeliveryDistanceKM (e.g., 3.8). **String / Text** ProductName ("Smartphone"), ProductCategory ("Electronics") Unicode character text strings used for names, categories, labels, and status codes. **Paytm:** PaymentStatus (e.g., "SUCCESS", "PENDING") or UPI_ID (e.g., "user@paytm").



DAX_Data_Types_Summary.xlsx

Session 15:

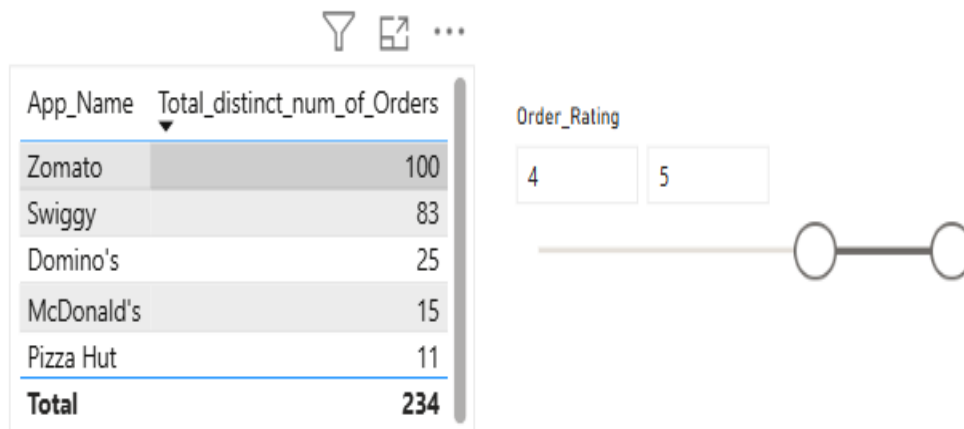
Task 1:

2.39M
Total_Orders

Task 2:

4.02
average_rating

Task 3:



Task 4:

1

2

3

4

5

6

7

8

9

10

11

12

new_Column =

VAR AverageOrderCount = [avgOrderCount]

RETURN

SWITCH(

TRUE(),

AppMetrics[appOrderCount] > AverageOrderCount, "High Performer", "Low"

)

App_ID	App_Name	Order_Count	Average_Rating	new_Column
APP-301	Swiggy	1520	4.2	Low
APP-302	Zomato	1850	4.5	Low
APP-303	Zepto	920	4.1	Low
APP-304	Blinkit	1410	4.4	Low
APP-305	Instamart	600	3.9	Low
APP-306	BigBasket	1100	4.3	Low
APP-307	Dunzo	450	3.6	Low
APP-308	EatSure	310	4	Low

Order_ID	App_Name	Bill_Amount_INR	User_Rating	TC
8052	Zomato	934	5	Top Choice
8072	Zomato	451	5	Top Choice
8064	Zomato	708	5	Top Choice
8003	Zomato	812	5	Top Choice
8117	Zomato	1098	5	Top Choice
8107	Zomato	962	5	Top Choice
8033	Zomato	741	5	Top Choice
8067	Zomato	711	5	Top Choice
8013	Zomato	601	5	Top Choice
8066	Zomato	835	5	Top Choice

Session 16:

Task 1:

Task 4:

Use the **FORMAT()** function to display each order date in the format 'dd-MMM-yyyy, dddd' (e.g., '05-Jun-2023, Monday') in a new column.

ANSWERS:

format
10-Jul-2025,10
25-Jul-2025,25
25-Aug-2025,25
05-Sep-2025,05
26-Sep-2025,26
15-Oct-2025,15
24-Oct-2025,24
02-Nov-2025,02
28-Dec-2025,28
02-Feb-2026,02
08-Mar-2026,08
12-Mar-2026,12
13-Mar-2026,13
22-Mar-2026,22
05-Jun-2026,05

Task 5:

Given a column of product codes like 'MOB1234IN', use the **LEFT()** and **RIGHT()** functions to extract the first three characters (category code) and the last two characters (region code) into separate columns.

Constraint: Do not use any custom M code — only DAX functions.

ANSWERS:

Product_ID	Product_Name	Category	Price	Discount	Quantity_Sold	LEFT	Right
PROD-10001	('iPhone 15 Pro', 'Electronics')	Electronics	84310	7031.4	199	PRO	01
PROD-10002	('Samsung Galaxy S24', 'Electronics')	Electronics	36548	4512.49	45	PRO	02
PROD-10003	Sony WH-1000XM5	Electronics	13934	3525.44	238	PRO	03
PROD-10004	Boat Nirvanaa	Electronics	71982	5476.51	118	PRO	04

Session 17:

Task 1:

10
Count_Rating

Task 2:

1.29K
Calculate_measures

Task 3:

Movie_Title	movies_collection_total	Rank_by_title
Animal	913	4
Baahubali 2	535	11
Bhool Bhulaiyaa 3	843	5
Dangal	1078	1
Gadar 2	483	13
Jawan	689	8
Kalki 2898 AD	713	6
KGF Chapter 2	554	10
Pathaan	504	12
Pushpa 2: The Rule	557	9
RRR	436	14
Sanju	388	15
Singham Again	692	7
Stree 2	1046	2
Tiger 3	1000	3
Total	10431	1

Task 4:

Use the SUMX() function to calculate the total value of items in a Myntra shopping cart table, where each item has a quantity and price column.

Hint: SUMX() lets you multiply quantity by price for each row and then sum the results.

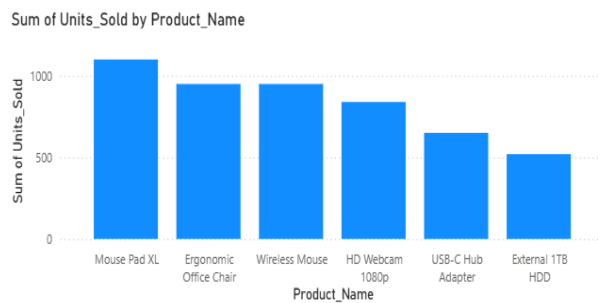
Category	cart_tbl
Footwear	33263
Apparel	30366
Accessories	13828
Total	77457

Task 5:

Restaurant_Name	avg_rating_per_resturant
Paradise Biryani	4.50
Truffles	4.40
Glen's Bakehouse	4.14
Barbeque Nation	4.00
Toit	3.83
Empire Restaurant	3.57
Social	3.14
Total	3.88

Session 18:

Task 1:



Task 2:

City	Payment_Method	Sum of Amount_INR
Ahmedabad	Net Banking	8159
Ahmedabad	UPI	4110
Total		12269

Task 3:

City	Payment_Method
Bangalore	Credit Card
Chennai	Credit Card
Delhi	Credit Card
Mumbai	Credit Card

Payment_Method

Credit Card ^

☐ Select all

☒ Credit Card

☐ Debit Card

☐ Net Banking

☐ UPI

Task 4:

→

←

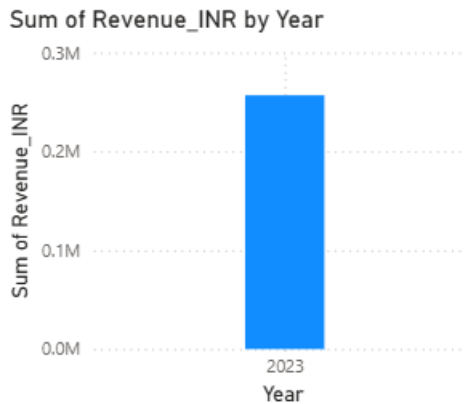
OverView_Page

Top_Customers

+

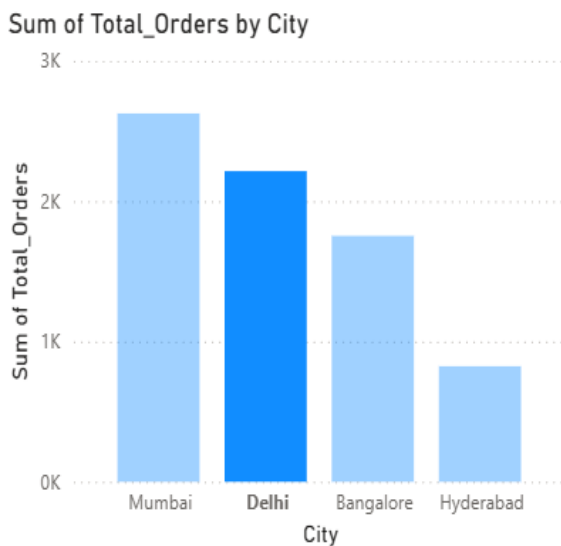
Task 5: Set up a Report Level Filter to show only data from the year 2023 throughout your entire Power BI report, regardless of the page or visual.

ANSWERS:



Session 19:

Task 1:



Restaurant_Name	Cuisine	Sum of Average_Rating
Burger Junction	Fast Food	3.70
Punjab Grill	North Indian	3.50
Waffle House	Desserts	3.30
Total		10.50

Task 2:



Task : 3

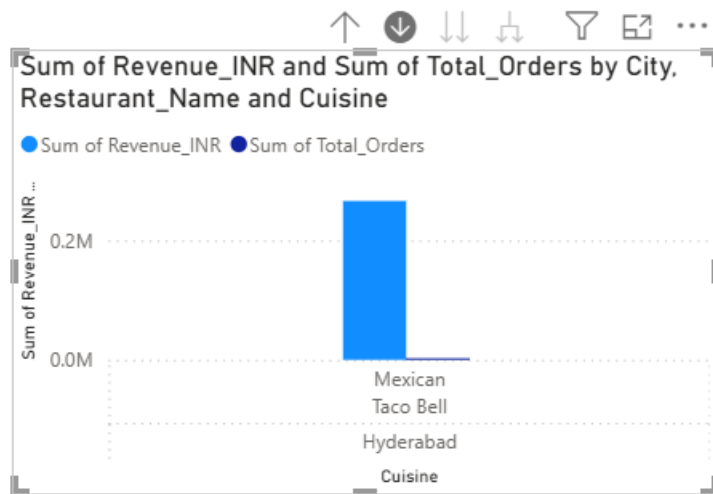
⬅

2.43K

Average of Price_INR

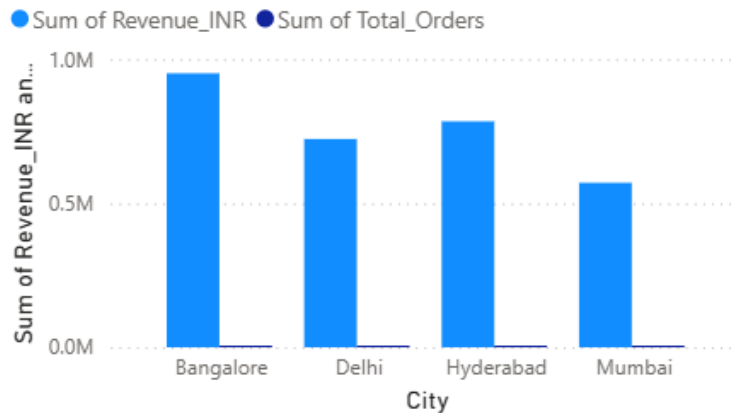
Sub_Category	Product_Name	Sum of Price_INR	Sum of Rating
Face Washes	Cetaphil Gentle Cleanser	3533	8.30
Face Washes	Garnier Skin Naturals	2467	4.00
Face Washes	Himalaya Purifying Neem	3885	4.00
Face Washes	Nivea Men Dark Spot	6198	11.10
Lipsticks	Lakme Absolute	8238	8.50
Lipsticks	MAC Retro Matte	6949	7.40
Lipsticks	Maybelline Color Sensation	3327	4.00
Total		72938	121.50

Task 4:



Task 5:

Sum of Revenue_INR and Sum of Total_Orders by City



Hierarchical Drill-Down Explanation

- **The Hierarchy Setup:** This visual contains a multi-level data hierarchy structured from **City** $\rightarrow$ **Restaurant_Name** $\rightarrow$ **Cuisine** stacked on the X-axis.
- **Drill-Down Behavior:** The image shows the chart in its fully expanded drill-down state. By using the navigation arrows at the top of the visual (specifically the fork-like expand icon $\rightarrow$), the chart dynamically navigates from a high-level summary down to the lowest grain.
- **The Result:** As a result of this deep navigation, the chart combines all levels together into a single view—displaying

Task 6:

Use ChatGPT or Copilot to generate step-by-step instructions for creating a drill-through page in Power BI, then follow those steps using your own dataset (not the one shown in class). After completing, compare the AI's instructions to your actual process and note any differences.

Hint Paste the AI's instructions in a text box on your report for reference.

ANSWERS:

Comparison: AI Instructions vs. My Actual Process

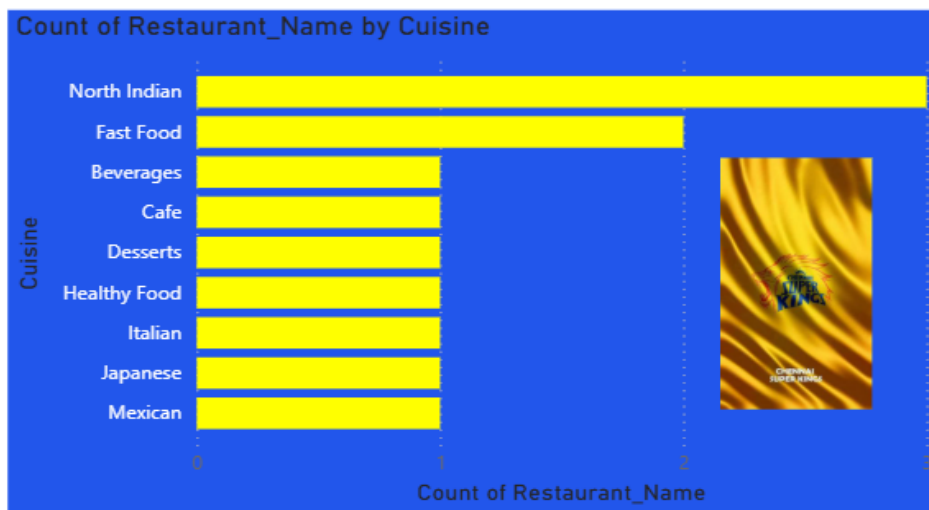
- **What the AI Instructions Said:** Look for the "Drill-through" section at the bottom of the *Visualizations* pane and drag your category field there.

- **What I Actually Did (The Difference):** In the latest version of Power BI Desktop, the Drill-through bucket has moved from the old Visualizations pane. I found it located at the bottom of the Filters pane under "Page-level filters" / "Drag drill-through fields here".
- **Hierarchical Difference:** The AI instructions assumed a simple, single-level click. Because my Zomato visual uses an advanced multi-level hierarchy (City > Restaurant > Cuisine), right-clicking on "Hyderabad" passed the entire drill-down path context over to the details page dynamically, which worked much better than the basic AI example.

Session 20:

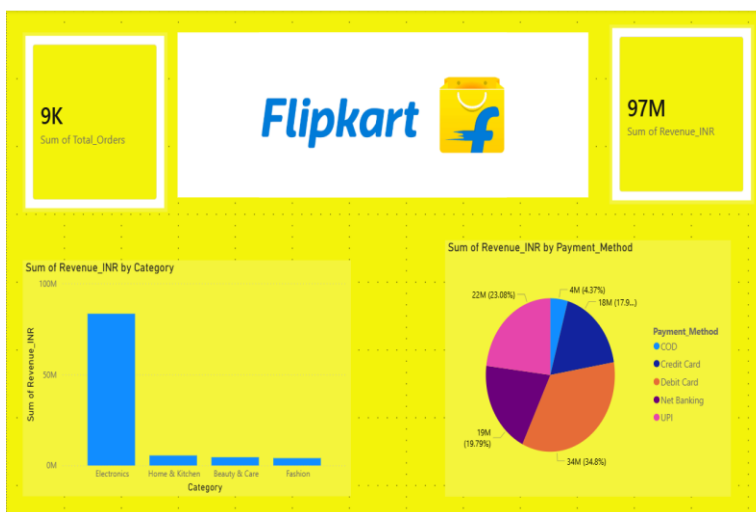
Task 1:

- 1.) Apply a custom color theme to a Power BI report that matches the branding of your favorite IPL team, including logo placement and color palette.



Task 2:

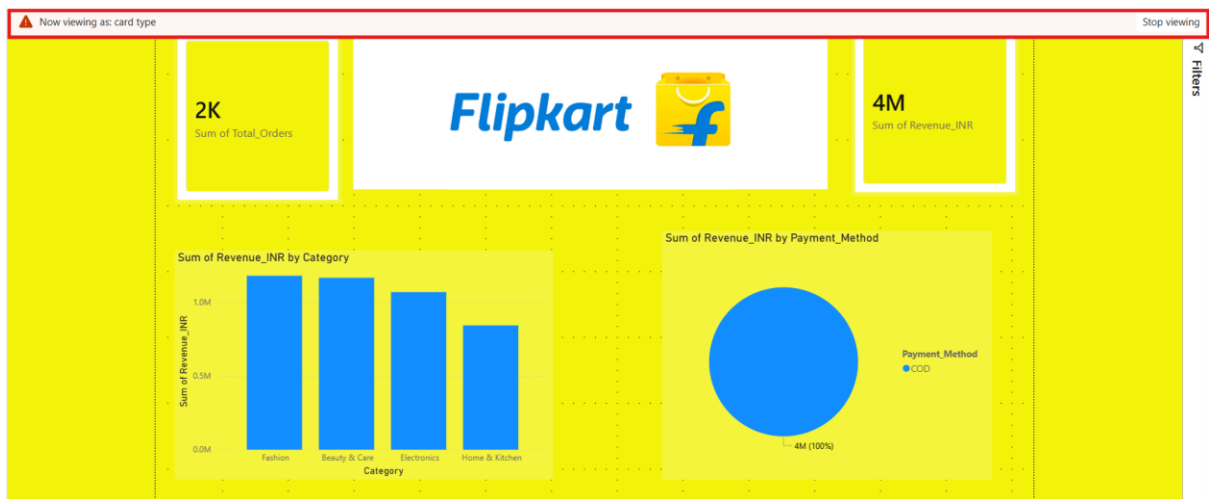
Design a Flipkart-style sales dashboard in Power BI using gridlines for alignment, and arrange visuals (bar chart, pie chart, KPI card) to create a clean, balanced layout.



Task 3:

Set up Row Level Security (RLS) in a Power BI report so that only sales data for a specific city (e.g., Ahmedabad) is visible to users from that city.

Hint: Use the 'Manage Roles' feature to define your rule and test it using 'View as Role'.



Task 4:

Publish your customized Power BI report to the Power BI Service and share it with a classmate using their email address.

Constraint: Ensure the report is accessible and viewable by the recipient.



Publish succeeded

Your report 'Flipkart_Sales_Dashboard.pbix' has been safely uploaded to the Power BI Cloud Service environment.

[Open 'Flipkart_Sales_Dashboard.pbix' in Power BI](#)

Task 5:

Align all visuals in your Power BI report using the alignment tools so that the dashboard looks professional and organized, similar to the clean layouts found in apps like Zomato or Spotify.

